COMPETENCE IN CABLE TECHNOLOGY

KERPEN Industrial



Instrumentation Cables

to Kerpen Standard

KERPEN

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2, group 2, classified areas (IEC 79-14), not allowed for direct connection to low impedance sources, e.g. public mains electricity supply

Not recommended for direct burial. For indoor and outdoor installation, in dry and wet locations; on racks, trays, in conduits.

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

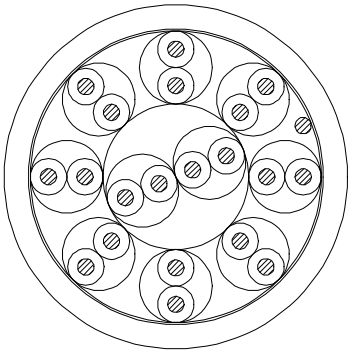
Colour code..... black / white;
continuously numbered on white cores 1-1,2-2 ... for multipairs

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK ˣ INSTRUMENTATION CABLE ˣ 300 V**
IEC 60332-2 part 3



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 (Cat.C)
Outer sheath:	
Amount of halogen acid gas:	**max. 17%** (IEC 754-1)
Limiting Oxygen Index (LOI):	**min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 300°C** (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	outer sheath of PVC
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor sizes	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39,2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1kHz	max.	nF/km					
single pair:			120				
up to 4 Pairs			96				
above 4 pairs			80				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage $U_{r.m.s.}$ ***core : core***		V	2000				
$U_{r.m.s.}$ ***core : screen***		V	1000				
Operating voltage $U_{r.m.s.}$	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.0	5.7	50	7279101S
2	0.4	1.0	8.1	60	7279102S
4	0.4	1.0	9.3	100	7279103S
6	0.4	1.0	11.0	140	7279104S
8	0.4	1.0	11.7	160	7279105S
10	0.4	1.2	13.6	200	7279106S
12	0.4	1.2	14.2	230	7279107S
16	0.4	1.2	16.0	290	7279108S
20	0.4	1.4	17.9	360	7279109S
24	0.4	1.4	19.3	420	7279110S
0.75 mm²					
1	0.4	1.0	6.1	60	7279111S
2	0.4	1.0	8.8	80	7279112S
4	0.4	1.0	10.1	120	7279113S
6	0.4	1.0	12.0	170	7279114S
8	0.4	1.2	13.2	220	7279115S
10	0.4	1.2	14.9	260	7279116S
12	0.4	1.2	15.6	300	7279117S
16	0.4	1.4	18.0	400	7279118S
20	0.4	1.4	19.8	480	7279119S
24	0.4	1.4	21.3	570	7279120S
1.0 mm²					
1	0.4	1.0	6.5	50	7279121S
2	0.4	1.0	9.4	100	7279122S
4	0.4	1.0	10.9	140	7279123S
6	0.4	1.2	13.4	220	7279124S
8	0.4	1.2	14.2	260	7279125S
10	0.4	1.2	16.1	320	7279126S
12	0.4	1.2	16.8	370	7279127S
16	0.4	1.4	19.4	500	7279128S
20	0.4	1.4	21.3	601	7279129S
24	0.4	1.6	23.5	730	7279130S

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
1.3 mm²					
1	0.4	1.0	6.8	70	7279131S
2	0.4	1.0	10.0	113	7279132S
4	0.4	1.0	11.6	170	7279133S
6	0.4	1.2	14.3	260	7279134S
8	0.4	1.2	15.2	320	7279135S
10	0.4	1.2	17.2	410	7279136S
12	0.4	1.4	18.4	480	7279137S
16	0.4	1.4	20.8	600	7279138S
20	0.4	1.6	23.3	780	7279139S
24	0.4	1.6	25.3	890	7279140S
1.5 mm²					
1	0.5	1.0	7.5	90	7279141S
2	0.5	1.0	11.1	130	7279142S
4	0.5	1.2	13.3	220	7279143S
6	0.5	1.2	15.9	310	7279144S
8	0.5	1.2	16.9	380	7279145S
10	0.5	1.4	19.7	480	7279146S
12	0.5	1.4	20.6	560	7279147S
16	0.5	1.6	23.8	740	7279148S
20	0.5	1.6	26.2	890	7279149S
24	0.5	1.8	28.7	1080	7279150S

RT = radial thickness

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2, group 2, classified areas (IEC 79-14), not allowed for direct connection to low impedance sources, e.g. public mains electricity supply

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations, on racks, in conduits

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white; continuously numbered on white cores 1-1, 2-2, ...)

Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

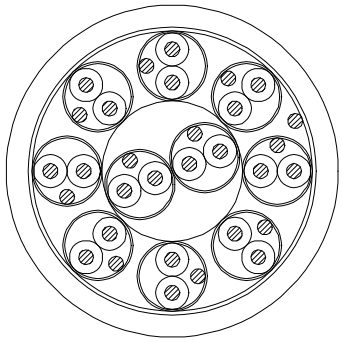
Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath.....polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK ˣ INSTRUMENTATION CABLE ˣ 300 V**

IEC 60332 PART 3



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 (Cat. C)

Outer sheath:

Amount of halogen acid gas: max. 17% (IEC 754-1)

Limiting Oxygen Index (LOI): min. 30% (IEC 60332-3 annex B)

Flame Temperature (FT): min. 300°C (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	outer sheath of PVC
-fl	reduced flame propagation
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor sizes	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	115				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage $U_{r.m.s.}$ ***core : core***		V	2000				
$U_{r.m.s.}$ ***core : screen***		V	1000				
Operating voltage $U_{r.m.s.}$	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl PiMF

Geometrical data					
No. of pairs	**RT of insulation nom. (mm)**	**RT of outer sheath nom. (mm)**	**Overall Diameter approx. (mm)**	**Weight approx. (kg/km)**	**Part No.**
0.5 mm²					
2	0.4	1.0	9.1	80	7279201S
4	0.4	1.0	10.5	120	7279202S
6	0.4	1.2	12.9	180	7279203S
8	0.4	1.2	13.7	210	7279204S
10	0.4	1.2	15.5	260	7279205S
12	0.4	1.2	16.2	290	7279206S
16	0.4	1.4	18.7	390	7279207S
20	0.4	1.4	20.5	460	7279208S
24	0.4	1.4	22.2	540	7279209S
0.75 mm²					
2	0.4	1.0	9.8	100	7279210S
4	0.4	1.0	11.4	140	7279211S
6	0.4	1.2	14.0	220	7279212S
8	0.4	1.2	14.9	260	7279213S
10	0.4	1.2	16.8	320	7279214S
12	0.4	1.4	18.0	380	7279215S
16	0.4	1.4	20.3	480	7279216S
20	0.4	1.4	22.4	580	7279217S
24	0.4	1.6	24.7	720	7279218S
1.0 mm²					
2	0.4	1.0	10.4	110	7279219S
4	0.4	1.2	12.5	180	7279220S
6	0.4	1.2	14.9	260	7279221S
8	0.4	1.2	15.8	310	7279222S
10	0.4	1.4	18.4	400	7279223S
12	0.4	1.4	19.2	451	7279224S
16	0.4	1.4	21.7	580	7279225S
20	0.4	1.6	24.4	730	7279226S
24	0.4	1.6	26.4	860	7279227S

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
1.3 mm²					
2	0.4	1.0	11.0	130	7279228S
4	0.4	1.2	13.2	210	7279229S
6	0.4	1.2	15.8	290	7279230S
8	0.4	1.2	16.8	370	7279231S
10	0.4	1.4	19.5	460	7279232S
12	0.4	1.4	20.5	530	7279233S
16	0.4	1.6	23.5	720	7279234S
20	0.4	1.6	26.0	930	7279235S
24	0.4	1.6	28.2	1030	7279236S
1.5 mm²					
2	0.5	1.2	12.5	150	7279237S
4	0.5	1.2	14.5	240	7279238S
6	0.5	1.2	17.4	320	7279239S
8	0.5	1.4	19.0	420	7279240S
10	0.5	1.4	21.6	510	7279241S
12	0.5	1.4	22.6	600	7279242S
16	0.5	1.6	26.1	810	7279243S
20	0.5	1.8	29.2	1030	7279244S
24	0.5	1.8	31.6	1220	7279245S

RT = radial thickness

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2, group 2, classified areas (IEC 79-14), not allowed for direct connection to low impedance sources, e.g. public mains electricity supply

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations, on racks, in conduits;

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

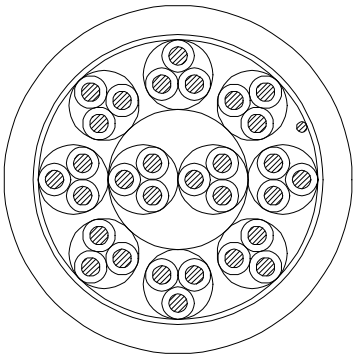
Colour code..... black / white / red;
continuously numbered on white cores (1-1-1, 2-2-2, ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK x INSTRUMENTATION CABLE x 300 V x IEC 60332 Part 3**



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 (Cat. C)
Outer sheath:	
Amount of halogen acid gas:	max. 17% (IEC 60754-1)
Limiting Oxygen Index (LOI):	min. 30% (IEC 60332-3 annex B)
Flame Temperature (FT):	min. 300°C (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	outer sheath of PVC
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor sizes	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triple:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage $U_{r.m.s.}$ ***core : core***		V	2000				
$U_{r.m.s.}$ ***core : screen***		V	1000				
Operating voltage $U_{r.m.s.}$	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.0	6.0		
2	0.4	1.0	8.9		
4	0.4	1.0	10.1		
6	0.4	1.2	12.9		
8	0.4	1.2	13.8		
10	0.4	1.2	15.5		
12	0.4	1.2	16.0		
16	0.4	1.4	18.4		
20	0.4	1.4	20.2		
24	0.4	1.4	21.8		
0.75 mm²					
1	0.4	1.0	6.4		
2	0.4	1.0	9.7		
4	0.4	1.0	11.1		
6	0.4	1.2	14.2		
8	0.4	1.2	15.1		
10	0.4	1.2	17.0		
12	0.4	1.4	18.0		
16	0.4	1.4	20.3		
20	0.4	1.4	22.3		
24	0.4	1.6	24.7		
1.0 mm²					
1	0.4	1.0	6.8		
2	0.4	1.0	10.4		
4	0.4	1.0	11.9		
6	0.4	1.2	15.2		
8	0.4	1.2	16.3		
10	0.4	1.4	18.8		
12	0.4	1.4	19.4		
16	0.4	1.4	21.9		
20	0.4	1.6	24.6		
24	0.4	1.6	26.6		

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl

1.3 mm²					
1	0.4	1.0	7.2		
2	0.4	1.0	11.1		
4	0.4	1.2	13.1		
6	0.4	1.2	16.3		
8	0.4	1.4	17.9		
10	0.4	1.4	20.1		
12	0.4	1.4	20.8		
16	0.4	1.6	24.0		
20	0.4	1.6	26.4		
24	0.4	1.8	29.0		
1.5 mm²					
1	0.5	1.0	7.9		
2	0.5	1.2	12.7		
4	0.5	1.2	14.6		
6	0.5	1.4	18.6		
8	0.5	1.4	19.9		
10	0.5	1.4	22.5		
12	0.5	1.6	23.8		
16	0.5	1.6	26.9		
20	0.5	1.8	30.1		
24	0.5	1.8	32.5		

RT = radial thickness

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2, group 2, classified areas (IEC 79-14), not allowed for direct connection to low impedance sources, e.g. public mains electricity supply

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations, on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red; continuously numbered on white cores (1-1-1, 2-2-2, ...)

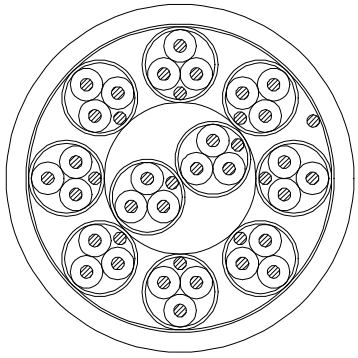
Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK x INSTRUMENTATION CABLE x 300 V x IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 (Cat. C)

Outer sheath:

Amount of halogen acid gas: max. 17% (IEC 60754-1)

Limiting Oxygen Index (LOI): min. 30% (IEC 60332-3 annex B)

Flame Temperature (FT): min. 300°C (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	outer sheath of PVC
-fl	reduced flame propagation
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor sizes	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	pF/250 m	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
2	0.4	1.0	10.0		
4	0.4	1.0	11.4		
6	0.4	1.2	14.6		
8	0.4	1.2	15.6		
10	0.4	1.4	17.9		
12	0.4	1.4	18.5		
16	0.4	1.4	20.9		
20	0.4	1.6	23.4		
24	0.4	1.6	25.4		
0.75 mm²					
2	0.4	1.0	10.8		
4	0.4	1.2	12.8		
6	0.4	1.2	15.8		
8	0.4	1.2	16.9		
10	0.4	1.4	19.5		
12	0.4	1.4	20.2		
16	0.4	1.4	22.8		
20	0.4	1.6	25.6		
24	0.4	1.6	27.7		
1.0 mm²					
2	0.4	1.0	11.5		
4	0.4	1.2	13.6		
6	0.4	1.2	16.9		
8	0.4	1.4	18.5		
10	0.4	1.4	20.9		
12	0.4	1.4	21.6		
16	0.4	1.6	24.9		
20	0.4	1.6	27.4		
24	0.4	1.8	30.1		

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Y-fl TiMF

1.3 mm²					
2	0.4	1.2	12.6		
4	0.4	1.2	14.4		
6	0.4	1.4	18.4		
8	0.4	1.4	19.7		
10	0.4	1.4	22.2		
12	0.4	1.6	23.4		
16	0.4	1.6	26.5		
20	0.4	1.8	29.6		
24	0.4	1.8	32.1		
1.5 mm²					
2	0.5	1.2	13.8		
4	0.5	1.2	15.9		
6	0.5	1.4	20.3		
8	0.5	1.4	21.7		
10	0.5	1.6	25.1		
12	0.5	1.6	26.0		
16	0.5	1.8	29.8		
20	0.5	1.8	32.9		
24	0.5	2.0	36.0		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 79 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national standards.For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

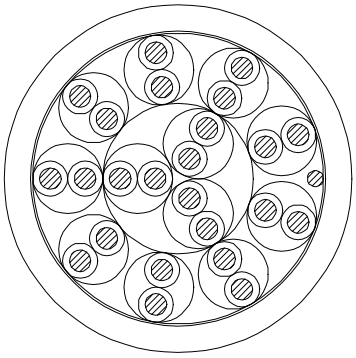
Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...) for multipairs

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... reinforced polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**
IEC 60332 PART 3



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Outer sheath:	
Amount of halogen acid gas:	**max. 17%** (IEC 754-1)
Limiting Oxygen Index (LOI):	**min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 300°C** (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Yv	outer sheath of PVC, reinforced
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.8	7.3	70	***7338052S***
2	0.4	1.8	9.7	100	***7338053S***
4	0.4	1.8	10.9	130	***7338054S***
6	0.4	1.8	12.6	180	7338055S
8	0.4	1.8	13.3	210	***7338056S***
10	0.4	1.8	14.8	240	7338057S
12	0.4	1.8	15.4	270	***7338058S***
16	0.4	1.8	17.2	340	***7338059S***
20	0.4	1.8	18.7	400	7338060S
24	0.4	1.8	20.1	460	***7338061S***
0.75 mm²					
1	0.4	1.8	7.7	80	***7338098S***
2	0.4	1.8	10.4	120	***7338099S***
4	0.4	1.8	11.7	160	***7338100S***
6	0.4	1.8	13.6	220	7338101S
8	0.4	1.8	14.4	260	***7338102S***
10	0.4	1.8	16.1	310	7338103S
12	0.4	1.8	16.8	344	***7338104S***
16	0.4	1.8	18.8	430	***7338105S***
20	0.4	1.8	20.6	520	7338106S
24	0.4	1.8	22.1	610	***7338107S***
1.0 mm²					
1	0.4	1.8	8.1	80	7338144S
2	0.4	1.8	11.0	140	7338145S
4	0.4	1.8	12.5	190	7338146S
6	0.4	1.8	14.6	260	7338147S
8	0.4	1.8	15.4	310	7338148S
10	0.4	1.8	17.3	370	7338149S
12	0.4	1.8	18.0	420	7338150S
16	0.4	1.8	20.2	530	7338151S
20	0.4	1.8	22.1	640	7338152S
24	0.4	2.0	24.3	770	7338153S

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

1.3 mm²					
1	0.4	1.8	8.4	100	***7338190S***
2	0.4	1.8	11.6	150	***7338191S***
4	0.4	1.8	13.2	220	***7338192S***
6	0.4	1.8	15.5	300	7338193S
8	0.4	1.8	16.4	360	***7338194S***
10	0.4	1.8	18.4	440	7338195S
12	0.4	1.8	19.2	520	***7338196S***
16	0.4	1.8	21.6	640	***7338197S***
20	0.4	2.0	24.1	780	7338198S
24	0.4	2.0	26.1	940	***7338199S***
1.5 mm²					
1	0.5	1.8	9.1	120	7338236S
2	0.5	1.8	12.7	180	7338237S
4	0.5	1.8	14.5	260	7338238S
6	0.5	1.8	17.1	360	7338239S
8	0.5	1.8	18.1	430	7338240S
10	0.5	1.8	20.5	520	7338241S
12	0.5	1.8	21.4	590	7338242S
16	0.5	2.0	24.6	780	7338243S
20	0.5	2.0	27.0	950	7338244S
24	0.5	2.0	29.1	1110	7338245S

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 79 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national standards.For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...)

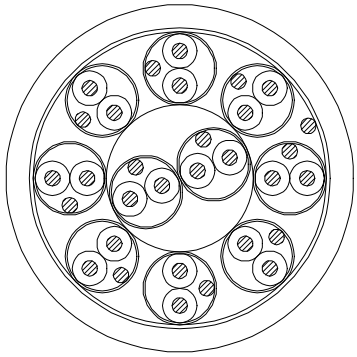
Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... reinforced polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**
IEC 60332 PART 3



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Outer sheath:	
Amount of halogen acid gas:	**max. 17%** (IEC 754-1)
Limiting Oxygen Index (LOI):	**min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 300°C** (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Yv	outer sheath of PVC, reinforced
-fl	reduced flame propagation
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
2	0.4	1.8	10.7	120	7338064S
4	0.4	1.8	12.1	160	7338065S
6	0.4	1.8	14.1	210	7338066S
8	0.4	1.8	14.9	250	7338067S
10	0.4	1.8	16.7	310	7338068S
12	0.4	1.8	17.4	340	7338069S
16	0.4	1.8	19.5	430	7338070S
20	0.4	1.8	21.3	500	7338071S
24	0.4	1.8	23.0	580	7338072S
0.75 mm²					
2	0.4	1.8	11.4	140	7338110S
4	0.4	1.8	13.0	190	7338111S
6	0.4	1.8	15.2	260	7338112S
8	0.4	1.8	16.1	310	7338113S
10	0.4	1.8	18.0	370	7338114S
12	0.4	1.8	18.8	410	7338115S
16	0.4	1.8	21.1	530	7338116S
20	0.4	1.8	23.2	640	7338117S
24	0.4	2.0	25.5	760	7338118S
1.0 mm²					
2	0.4	1.8	12.0	160	7338156S
4	0.4	1.8	13.7	220	7338157S
6	0.4	1.8	16.1	300	7338158S
8	0.4	1.8	17.0	360	7338159S
10	0.4	1.8	19.2	430	7338160S
12	0.4	1.8	20.0	490	7338161S
16	0.4	1.8	22.5	620	7338162S
20	0.4	2.0	25.2	780	7338163S
24	0.4	2.0	27.2	920	7338164S

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl PiMF

1.3 mm²					
2	0.4	1.8	12.6	170	7338202S
4	0.4	1.8	14.4	250	7338203S
6	0.4	1.8	17.0	350	7338204S
8	0.4	1.8	18.0	420	7338205S
10	0.4	1.8	20.3	500	7338206S
12	0.4	1.8	21.3	580	7338207S
16	0.4	2.0	24.3	760	7338208S
20	0.4	2.0	26.8	930	7338209S
24	0.4	2.0	29.0	1090	7338210S
1.5 mm²					
2	0.5	1.8	13.7	190	7338248S
4	0.5	1.8	15.7	290	7338249S
6	0.5	1.8	18.6	410	7338250S
8	0.5	1.8	19.8	500	7338251S
10	0.5	1.8	22.4	590	7338252S
12	0.5	1.8	23.4	680	7338253S
16	0.5	2.0	26.9	890	7338254S
20	0.5	2.0	29.6	1090	7338255S
24	0.5	2.0	32.0	1270	7338256S

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 79 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national standards.For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

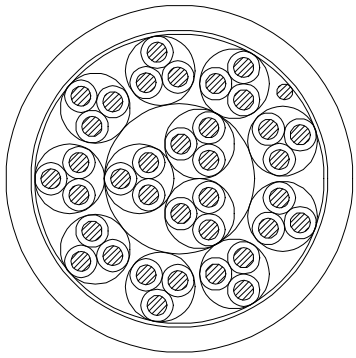
Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ..) for multitriples

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... reinforced polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**
IEC 60332 PART 3



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Outer sheath:	
Amount of halogen acid gas:	**max. 17%** (IEC 754-1)
Limiting Oxygen Index (LOI):	**min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 300°C** (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Yv	outer sheath of PVC, reinforced
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.8	7.6	80	7338075S
2	0.4	1.8	10.5	120	7338076S
4	0.4	1.8	11.7	160	7338077S
6	0.4	1.8	14.1	230	7338078S
8	0.4	1.8	15.0	270	7338079S
10	0.4	1.8	16.7	320	7338080S
12	0.4	1.8	17.2	360	7338081S
16	0.4	1.8	19.2	450	7338082S
20	0.4	1.8	21.0	540	7338083S
24	0.4	1.8	22.6	630	7338084S
0.75 mm²					
1	0.4	1.8	8.0	90	7338121S
2	0.4	1.8	11.3	140	7338122S
4	0.4	1.8	12.7	210	7338123S
6	0.4	1.8	15.4	290	7338124S
8	0.4	1.8	16.3	340	7338125S
10	0.4	1.8	18.2	420	7338126S
12	0.4	1.8	18.8	470	7338127S
16	0.4	1.8	21.1	600	7338128S
20	0.4	1.8	23.1	720	7338129S
24	0.4	2.0	25.5	850	7338130S
1.0 mm²					
1	0.4	1.8	8.4	90	7338167S
2	0.4	1.8	12.0	170	7338168S
4	0.4	1.8	13.5	250	7338169S
6	0.4	1.8	16.4	350	7338170S
8	0.4	1.8	17.5	420	7338171S
10	0.4	1.8	19.6	510	7338172S
12	0.4	1.8	20.2	580	7338173S
16	0.4	1.8	22.7	740	7338174S
20	0.4	2.0	25.4	910	7338175S
24	0.4	2.0	27.4	1060	7338176S

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl

1.3 mm²					
1	0.4	1.8	8.8	120	7338213S
2	0.4	1.8	12.7	190	7338214S
4	0.4	1.8	14.3	290	7338215S
6	0.4	1.8	17.5	420	7338216S
8	0.4	1.8	18.7	500	7338217S
10	0.4	1.8	20.9	610	7338218S
12	0.4	1.8	21.6	700	7338219S
16	0.4	2.0	24.8	940	7338220S
20	0.4	2.0	27.2	1130	7338221S
24	0.4	2.0	29.4	1320	7338222S
1.5 mm²					
1	0.5	1.8	9.5	140	7338259S
2	0.5	1.8	13.9	220	7338260S
4	0.5	1.8	15.8	340	7338261S
6	0.5	1.8	19.4	480	7338262S
8	0.5	1.8	20.7	600	7338263S
10	0.5	1.8	23.3	720	7338264S
12	0.5	2.0	24.6	860	7338265S
16	0.5	2.0	27.7	1110	7338266S
20	0.5	2.0	30.5	1340	7338267S
24	0.5	2.0	32.9	1570	7338268S

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl DiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 79 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national standards.For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, 7-strands, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ..)

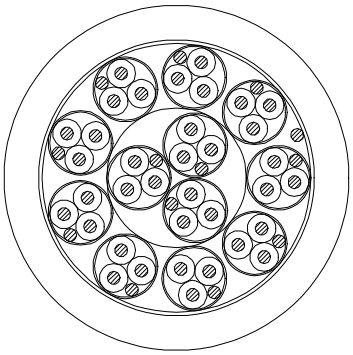
Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... reinforced polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**
IEC 60332 PART 3



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Outer sheath:	
Amount of halogen acid gas:	**max. 17%** (IEC 754-1)
Limiting Oxygen Index (LOI):	**min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 300°C** (ISO 4589-3)
Sunlight resistance:	(UL 1581 section 1200)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Yv	outer sheath of PVC, reinforced
-fl	reduced flame propagation
DiMF	triple screen

Electrical data at 20°C

	Character	***Unit***	***Values***				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl DiMF

Geometrical data					
No. of triples	**RT of insulation nom. (mm)**	**RT of outer sheath nom. (mm)**	**Overall Diameter approx. (mm)**	**Weight approx. (kg/km)**	**Part No.**
0.5 mm²					
2	0.4	1.8	11.6	140	7338087S
4	0.4	1.8	13.0	190	7338088S
6	0.4	1.8	15.8	270	7338089S
8	0.4	1.8	16.8	320	7338090S
10	0.4	1.8	18.7	390	7338091S
12	0.4	1.8	19.3	420	7338092S
16	0.4	1.8	21.7	540	7338093S
20	0.4	2.0	24.2	670	7338094S
24	0.4	2.0	26.2	800	7338095S
0.75 mm²					
2	0.4	1.8	12.4	170	7338133S
4	0.4	1.8	14.0	240	7338134S
6	0.4	1.8	17.0	340	7338135S
8	0.4	1.8	18.1	410	7338136S
10	0.4	1.8	20.3	490	7338137S
12	0.4	1.8	21.0	550	7338138S
16	0.4	1.8	23.6	720	7338139S
20	0.4	2.0	26.4	870	7338140S
24	0.4	2.0	28.5	1020	7338141S
1.0 mm²					
2	0.4	1.8	13.1	190	7338179S
4	0.4	1.8	14.8	280	7338180S
6	0.4	1.8	18.1	390	7338181S
8	0.4	1.8	19.3	480	7338182S
10	0.4	1.8	21.7	580	7338183S
12	0.4	1.8	22.4	660	7338184S
16	0.4	2.0	25.7	870	7338185S
20	0.4	2.0	28.2	1050	7338186S
24	0.4	2.0	30.5	1230	7338187S

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, PVC-Sheath **70°C / 300 V**

RE-2Y(St)Yv-fl DiMF

1.3 mm²					
2	0.4	1.8	13.8	210	7338225S
4	0.4	1.8	15.6	320	7338226S
6	0.4	1.8	19.2	460	7338227S
8	0.4	1.8	20.5	560	7338228S
10	0.4	1.8	23.0	680	7338229S
12	0.4	2.0	24.2	810	7338230S
16	0.4	2.0	27.3	1040	7338231S
20	0.4	2.0	30.0	1260	7338232S
24	0.4	2.0	32.5	1480	7338233S
1.5 mm²					
2	0.5	1.8	15.0	250	7338271S
4	0.5	1.8	17.1	380	7338272S
6	0.5	1.8	21.1	540	7338273S
8	0.5	1.8	22.5	650	7338274S
10	0.5	2.0	25.9	820	7338275S
12	0.5	2.0	26.8	950	7338276S
16	0.5	2.0	30.2	1210	7338277S
20	0.5	2.0	33.3	1480	7338278S
24	0.5	2.2	36.4	1770	7338279S

RT = radial thickness

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2, group 2, classified areas (IEC 79-14), not allowed for direct connection to low impedance sources, e.g. public mains electricity supply

Recommended for direct burial. For indoor and outdoor installation in dry and wet locations, on racks, in conduits

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white;
continuously numbered on white core (1-1, 2-2,...) for multipairs

Wrapping..... at least 1 layer of plastic tape

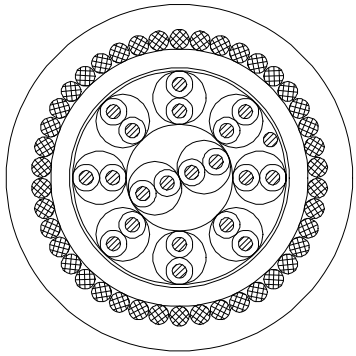
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK ᵡ INSTRUMENTATION CABLE ᵡ 300 V ᵡ IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: max. 17% (IEC 60754-1)

Limiting Oxygen Index (LOI): min. 30% (IEC 60332-3 annex B)

Flame Temperature (FT): min. 300°C (ISO 4589-3

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
SWA	steel wire armour
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	0.8	5.3	0.9	1.3	9.7		
2	0.4	0.9	7.9	0.9	1.4	12.5		
4	0.4	0.9	9.1	0.9	1.4	13.7		
6	0.4	1.1	11.2	0.9	1.5	16.0		
8	0.4	1.1	11.9	0.9	1.5	16.7		
10	0.4	1.2	13.6	1.25	1.6	19.3		
12	0.4	1.2	14.2	1.25	1.6	19.9		
16	0.4	1.2	16.0	1.25	1.6	21.7		
20	0.4	1.3	17.7	1.6	1.7	24.3		
24	0.4	1.3	19.1	1.6	1.7	25.7		
0.75 mm²								
1	0.4	0.8	5.7	0.9	1.3	10.1		
2	0.4	0.9	8.6	0.9	1.4	13.2		
4	0.4	1.1	10.3	0.9	1.4	14.9		
6	0.4	1.1	12.2	0.9	1.5	17.0		
8	0.4	1.2	13.2	1.25	1.5	18.7		
10	0.4	1.2	14.9	1.25	1.6	20.6		
12	0.4	1.2	15.6	1.25	1.6	21.3		
16	0.4	1.3	17.8	1.6	1.7	24.4		
20	0.4	1.3	19.6	1.6	1.7	26.2		
24	0.4	1.3	21.1	1.6	1.8	27.9		
1.0 mm²								
1	0.4	0.8	6.1	0.9	1.3	10.5		
2	0.4	0.9	9.2	0.9	1.4	13.8		
4	0.4	1.1	11.1	0.9	1.5	15.9		
6	0.4	1.2	13.4	1.25	1.5	18.9		
8	0.4	1.2	14.2	1.25	1.6	19.9		
10	0.4	1.2	16.1	1.25	1.6	21.8		
12	0.4	1.2	16.8	1.25	1.6	22.5		
16	0.4	1.3	19.2	1.6	1.7	25.8		
20	0.4	1.3	21.1	1.6	1.8	27.9		
24	0.4	1.5	23.3	1.6	1.8	30.1		

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Instrumentation Cable

KERPEN - STANDARD

Flame Retardant

Single- & Multi-Pair, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

1.3 mm²								
1	0.4	0.8	6.4	0.9	1.3	10.8		
2	0.4	0.9	9.8	0.9	1.4	14.4		
4	0.4	1.1	11.8	0.9	1.5	16.6		
6	0.4	1.2	14.3	1.25	1.6	20.0		
8	0.4	1.2	15.2	1.25	1.6	20.9		
10	0.4	1.2	17.2	1.25	1.7	23.1		
12	0.4	1.3	18.2	1.6	1.7	24.8		
16	0.4	1.3	20.6	1.6	1.7	27.2		
20	0.4	1.3	22.7	1.6	1.8	29.5		
24	0.4	1.5	25.1	1.6	1.9	32.1		
1.5 mm²								
1	0.5	0.8	7.1	0.9	1.3	11.5		
2	0.5	1.1	11.3	0.9	1.5	16.1		
4	0.5	1.2	13.3	1.25	1.5	18.8		
6	0.5	1.2	15.9	1.25	1.6	21.6		
8	0.5	1.2	16.9	1.25	1.7	22.8		
10	0.5	1.3	19.5	1.6	1.7	26.1		
12	0.5	1.3	20.4	1.6	1.7	27.0		
16	0.5	1.5	23.6	1.6	1.8	30.4		
20	0.5	1.5	26.0	1.6	1.9	33.0		
24	0.5	1.5	28.1	2.0	2.0	36.1		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white; continuously numbered on cores (1-1, 2-2, ...)

Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

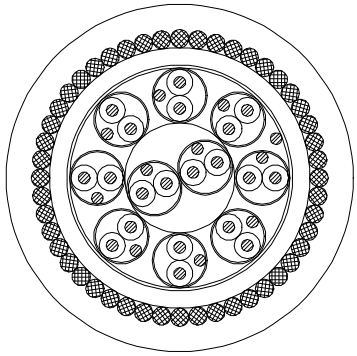
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
SWA	steel wire armour
-fl	reduced flame propagation
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	0.9	8.9	0.9	1.4	13.5		
4	0.4	1.1	10.7	0.9	1.4	15.3		
6	0.4	1.2	12.9	1.25	1.5	18.4		
8	0.4	1.2	13.7	1.25	1.6	19.4		
10	0.4	1.2	15.5	1.25	1.6	21.2		
12	0.4	1.2	16.2	1.25	1.6	21.9		
16	0.4	1.3	18.5	1.6	1.7	25.1		
20	0.4	1.3	20.3	1.6	1.7	26.9		
24	0.4	1.3	22.0	1.6	1.8	28.8		
0.75 mm²								
2	0.4	0.9	9.6	0.9	1.4	14.2		
4	0.4	1.1	11.6	0.9	1.5	16.4		
6	0.4	1.2	14.0	1.25	1.6	19.7		
8	0.4	1.2	14.9	1.25	1.6	20.6		
10	0.4	1.2	16.8	1.25	1.6	22.5		
12	0.4	1.3	17.8	1.6	1.7	24.4		
16	0.4	1.3	20.1	1.6	1.7	26.7		
20	0.4	1.3	22.2	1.6	1.8	29.0		
24	0.4	1.5	24.5	1.6	1.9	31.5		
1.0 mm²								
2	0.4	1.1	10.6	0.9	1.4	15.2		
4	0.4	1.2	12.5	1.25	1.5	18.0		
6	0.4	1.2	14.9	1.25	1.6	20.6		
8	0.4	1.2	15.8	1.25	1.6	21.5		
10	0.4	1.3	18.2	1.6	1.7	24.8		
12	0.4	1.3	19.0	1.6	1.7	25.6		
16	0.4	1.3	21.5	1.6	1.8	28.3		
20	0.4	1.5	24.2	1.6	1.9	31.2		
24	0.4	1.5	26.2	1.6	1.9	33.2		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl PiMF

1.3 mm²								
2	0.4	1.1	11.2	0.9	1.5	16.0		
4	0.4	1.2	13.2	1.25	1.5	18.7		
6	0.4	1.2	15.8	1.25	1.6	21.5		
8	0.4	1.2	16.8	1.25	1.6	22.5		
10	0.4	1.3	19.3	1.6	1.7	25.9		
12	0.4	1.3	20.3	1.6	1.7	26.9		
16	0.4	1.5	23.3	1.6	1.8	30.1		
20	0.4	1.5	25.8	1.6	1.9	32.8		
24	0.4	1.5	28.0	2.0	2.0	36.0		
1.5 mm²								
2	0.5	1.2	12.5	1.25	1.5	18.0		
4	0.5	1.2	14.5	1.25	1.6	20.2		
6	0.5	1.2	17.4	1.25	1.7	23.3		
8	0.5	1.3	18.8	1.6	1.7	25.4		
10	0.5	1.3	21.4	1.6	1.8	28.2		
12	0.5	1.3	22.4	1.6	1.8	29.2		
16	0.5	1.5	25.9	1.6	1.9	32.9		
20	0.5	1.7	29.0	2.0	2.0	37.0		
24	0.5	1.7	31.4	2.0	2.0	39.4		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

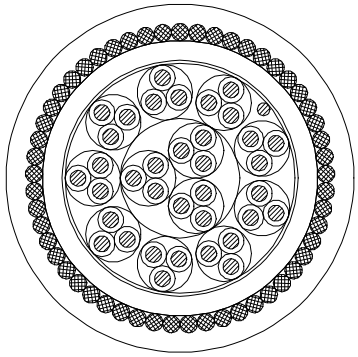
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**
IEC 60332 PART 3



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C

Outer sheath:
Amount of halogen acid gas: **max. 17%** (IEC 60754-1)
Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 300°C** (ISO 4589-3)
Sunlight resistance: (UL 1581 section 1200)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
SWA	steel wire armour
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	0.8	5.6	0.9	1.3	10.0		
2	0.4	0.9	8.7	0.9	1.4	13.3		
4	0.4	1.1	10.3	0.9	1.4	14.9		
6	0.4	1.2	12.9	1.25	1.5	18.4		
8	0.4	1.2	13.8	1.25	1.6	19.5		
10	0.4	1.2	15.5	1.25	1.6	21.2		
12	0.4	1.2	16.0	1.25	1.6	21.7		
16	0.4	1.3	18.2	1.6	1.7	24.8		
20	0.4	1.3	20.0	1.6	1.7	26.6		
24	0.4	1.3	21.6	1.6	1.8	28.4		
0.75 mm²								
1	0.4	0.8	6.0	0.9	1.3	10.4		
2	0.4	0.9	9.5	0.9	1.4	14.1		
4	0.4	1.1	11.3	0.9	1.5	16.1		
6	0.4	1.2	14.2	1.25	1.6	19.9		
8	0.4	1.2	15.1	1.25	1.6	20.8		
10	0.4	1.2	17.0	1.25	1.7	22.9		
12	0.4	1.3	17.8	1.6	1.7	24.4		
16	0.4	1.3	20.1	1.6	1.7	26.7		
20	0.4	1.3	22.1	1.6	1.8	28.9		
24	0.4	1.5	24.5	1.6	1.9	31.5		
1.0 mm²								
1	0.4	0.8	6.4	0.9	1.3	10.8		
2	0.4	1.1	10.6	0.9	1.4	15.2		
4	0.4	1.1	12.1	0.9	1.5	16.9		
6	0.4	1.2	15.2	1.25	1.6	20.9		
8	0.4	1.2	16.3	1.25	1.6	22.0		
10	0.4	1.3	18.6	1.6	1.7	25.2		
12	0.4	1.3	19.2	1.6	1.7	25.8		
16	0.4	1.3	21.7	1.6	1.8	28.5		
20	0.4	1.5	24.4	1.6	1.9	31.4		
24	0.4	1.5	26.4	1.6	1.9	33.4		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl

1.3 mm²								
1	0.4	0.8	6.8	0.9	1.3	11.2		
2	0.4	1.1	11.3	0.9	1.5	16.1		
4	0.4	1.2	13.1	1.25	1.5	18.6		
6	0.4	1.2	16.3	1.25	1.6	22.0		
8	0.4	1.3	17.7	1.6	1.7	24.3		
10	0.4	1.3	19.9	1.6	1.7	26.5		
12	0.4	1.3	20.6	1.6	1.7	27.2		
16	0.4	1.5	23.8	1.6	1.8	30.6		
20	0.4	1.5	26.2	1.6	1.9	33.2		
24	0.4	1.7	28.8	2.0	2.0	36.8		
1.5 mm²								
1	0.5	0.8	7.5	0.9	1.4	12.1		
2	0.5	1.2	12.7	1.25	1.5	18.2		
4	0.5	1.2	14.6	1.25	1.6	20.3		
6	0.5	1.3	18.4	1.6	1.7	25.0		
8	0.5	1.3	19.7	1.6	1.7	26.3		
10	0.5	1.3	22.3	1.6	1.8	29.1		
12	0.5	1.5	23.6	1.6	1.8	30.4		
16	0.5	1.5	26.7	1.6	1.9	33.7		
20	0.5	1.7	29.9	2.0	2.0	37.9		
24	0.5	1.7	32.3	2.0	2.1	40.5		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...)

Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

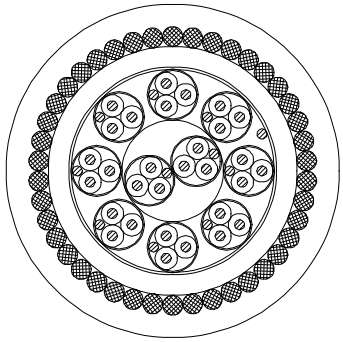
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
SWA	steel wire armour
-fl	reduced flame propagation
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	0.9	9.8	0.9	1.4	14.4		
4	0.4	1.1	11.6	0.9	1.5	16.4		
6	0.4	1.2	14.6	1.25	1.6	20.3		
8	0.4	1.2	15.6	1.25	1.6	21.3		
10	0.4	1.3	17.7	1.6	1.7	24.3		
12	0.4	1.3	18.3	1.6	1.7	24.9		
16	0.4	1.3	20.7	1.6	1.7	27.3		
20	0.4	1.3	22.8	1.6	1.8	29.6		
24	0.4	1.5	25.2	1.6	1.9	32.2		
0.75 mm²								
2	0.4	1.1	11.0	0.9	1.5	15.8		
4	0.4	1.2	12.8	1.25	1.5	18.3		
6	0.4	1.2	15.8	1.25	1.6	21.5		
8	0.4	1.2	16.9	1.25	1.7	22.8		
10	0.4	1.3	19.3	1.6	1.7	25.9		
12	0.4	1.3	20.0	1.6	1.7	26.6		
16	0.4	1.3	22.6	1.6	1.8	29.4		
20	0.4	1.5	25.4	1.6	1.9	32.4		
24	0.4	1.5	27.5	1.6	1.9	34.5		
1.0 mm²								
2	0.4	1.1	11.7	0.9	1.5	16.5		
4	0.4	1.2	13.6	1.25	1.6	19.3		
6	0.4	1.2	16.9	1.25	1.7	22.8		
8	0.4	1.3	18.3	1.6	1.7	24.9		
10	0.4	1.3	20.7	1.6	1.7	27.3		
12	0.4	1.3	21.4	1.6	1.8	28.2		
16	0.4	1.5	24.7	1.6	1.9	31.7		
20	0.4	1.5	27.2	1.6	1.9	34.2		
24	0.4	1.7	29.9	2.0	2.0	37.9		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YSWAY-fl TiMF

1.3 mm²								
2	0.4	1.2	12.6	1.25	1.5	18.1		
4	0.4	1.2	14.4	1.25	1.6	20.1		
6	0.4	1.3	18.2	1.6	1.7	24.8		
8	0.4	1.3	19.5	1.6	1.7	26.1		
10	0.4	1.3	22.0	1.6	1.8	28.8		
12	0.4	1.3	22.8	1.6	1.8	29.6		
16	0.4	1.5	26.3	1.6	1.9	33.3		
20	0.4	1.7	29.4	2.0	2.0	37.4		
24	0.4	1.7	31.9	2.0	2.1	40.1		
1.5 mm²								
2	0.5	1.2	13.8	1.25	1.6	19.5		
4	0.5	1.2	15.9	1.25	1.6	21.6		
6	0.5	1.3	20.1	1.6	1.7	26.7		
8	0.5	1.3	21.5	1.6	1.8	28.3		
10	0.5	1.5	24.9	1.6	1.9	31.9		
12	0.5	1.5	25.8	1.6	1.9	32.8		
16	0.5	1.7	29.6	2.0	2.0	37.6		
20	0.5	1.7	32.7	2.0	2.1	40.9		
24	0.5	2.0	36.0	2.0	2.2	44.4		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white; continuously numbered on cores (1-1, 2-2, ...) for multipairs

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

Metal sheath..... lead

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
15 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
M	sheath of lead
SWA	steel wire armour
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	RT of lead sheath nom. (mm)	Dia. over lead sheath approx. (mm)	RT of bedding nom. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²										
1	0.4	0.8	1.1	8.1	0.8	0.9	1.4	14.3		
2	0.4	0.9	1.1	10.7	0.8	0.9	1.5	17.1		
4	0.4	0.9	1.1	11.9	0.8	1.25	1.5	19.0		
6	0.4	1.1	1.1	14.0	0.8	1.25	1.6	21.3		
8	0.4	1.2	1.1	14.9	0.8	1.25	1.6	22.2		
10	0.4	1.2	1.1	16.4	1.0	1.6	1.7	25.0		
12	0.4	1.2	1.2	17.2	1.0	1.6	1.7	25.8		
16	0.4	1.2	1.2	19.0	1.0	1.6	1.8	27.8		
20	0.4	1.3	1.3	20.9	1.0	1.6	1.8	29.7		
24	0.4	1.3	1.3	22.3	1.0	1.6	1.9	31.3		
0.75 mm²										
1	0.4	0.8	1.1	8.5	0.8	0.9	1.4	14.7		
2	0.4	0.9	1.1	11.4	0.8	1.25	1.5	18.5		
4	0.4	1.1	1.1	13.1	0.8	1.25	1.6	20.4		
6	0.4	1.2	1.1	15.2	1.0	1.25	1.7	23.1		
8	0.4	1.2	1.1	16.0	1.0	1.6	1.7	24.6		
10	0.4	1.2	1.2	17.9	1.0	1.6	1.7	26.5		
12	0.4	1.2	1.2	18.6	1.0	1.6	1.7	27.2		
16	0.4	1.3	1.3	21.0	1.0	1.6	1.8	29.8		
20	0.4	1.3	1.3	22.8	1.0	1.6	1.9	31.8		
24	0.4	1.3	1.3	24.3	1.0	1.6	1.9	33.3		
1.0 mm²										
1	0.4	0.8	1.1	8.9	0.8	0.9	1.4	15.1		
2	0.4	0.9	1.1	12.0	0.8	1.25	1.6	19.3		
4	0.4	1.1	1.1	13.9	0.8	1.25	1.6	21.2		
6	0.4	1.2	1.1	16.2	1.0	1.6	1.7	24.8		
8	0.4	1.2	1.2	17.2	1.0	1.6	1.7	25.8		
10	0.4	1.2	1.2	19.1	1.0	1.6	1.8	27.9		
12	0.4	1.2	1.2	19.8	1.0	1.6	1.8	28.6		
16	0.4	1.3	1.3	22.4	1.0	1.6	1.9	31.4		
20	0.4	1.3	1.3	24.3	1.0	1.6	1.9	33.3		
24	0.4	1.5	1.4	26.7	1.2	2.0	2.0	37.1		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

1.3 mm²										
1	0.4	0.8	1.1	9.2	0.8	0.9	1.4	15.4		
2	0.4	1.1	1.1	13.0	0.8	1.25	1.6	20.3		
4	0.4	1.2	1.1	14.8	0.8	1.25	1.6	22.1		
6	0.4	1.2	1.2	17.3	1.0	1.6	1.7	25.9		
8	0.4	1.2	1.2	18.2	1.0	1.6	1.7	26.8		
10	0.4	1.3	1.2	20.4	1.0	1.6	1.8	29.2		
12	0.4	1.3	1.3	21.4	1.0	1.6	1.8	30.2		
16	0.4	1.3	1.3	23.8	1.0	1.6	1.9	32.8		
20	0.4	1.5	1.4	26.5	1.2	2.0	2.0	36.9		
24	0.4	1.5	1.5	28.6	1.2	2.0	2.0	39.0		
1.5 mm²										
1	0.5	0.8	1.1	9.9	0.8	0.9	1.5	16.3		
2	0.5	1.1	1.1	14.1	0.8	1.25	1.6	21.4		
4	0.5	1.2	1.1	16.1	1.0	1.6	1.7	24.7		
6	0.5	1.2	1.2	18.9	1.0	1.6	1.7	27.5		
8	0.5	1.2	1.2	19.9	1.0	1.6	1.8	28.7		
10	0.5	1.3	1.3	22.7	1.0	1.6	1.9	31.7		
12	0.5	1.3	1.3	23.6	1.0	1.6	1.9	32.6		
16	0.5	1.5	1.4	26.9	1.2	2.0	2.0	37.3		
20	0.5	1.5	1.5	29.5	1.2	2.0	2.1	40.1		
24	0.5	1.7	1.6	32.2	1.2	2.0	2.2	43.0		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white; continuously numbered on cores (1-1, 2-2, ...)

Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

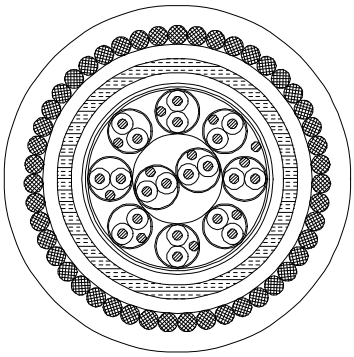
Metal sheath..... lead

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
15 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
M	sheath of lead
SWA	steel wire armour
-fl	reduced flame propagation
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	RT of lead sheath nom. (mm)	Dia. over lead sheath approx. (mm)	RT of bedding nom. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²										
2	0.4	0.9	1.1	11.7	0.8	1.25	1.5	18.8		
4	0.4	1.1	1.1	13.5	0.8	1.25	1.6	20.8		
6	0.4	1.2	1.1	15.7	1.0	1.6	1.7	24.3		
8	0.4	1.2	1.1	16.5	1.0	1.6	1.7	25.1		
10	0.4	1.2	1.2	18.5	1.0	1.6	1.7	27.1		
12	0.4	1.2	1.2	19.2	1.0	1.6	1.8	28.0		
16	0.4	1.3	1.3	21.7	1.0	1.6	1.8	30.5		
20	0.4	1.3	1.3	23.5	1.0	1.6	1.9	32.5		
24	0.4	1.3	1.4	25.4	1.2	1.6	1.9	34.8		
0.75 mm²										
2	0.4	1.1	1.1	12.8	0.8	1.25	1.6	20.1		
4	0.4	1.1	1.1	14.4	0.8	1.25	1.6	21.7		
6	0.4	1.2	1.1	16.8	1.0	1.6	1.7	25.4		
8	0.4	1.2	1.2	17.9	1.0	1.6	1.7	26.5		
10	0.4	1.2	1.2	19.8	1.0	1.6	1.8	28.6		
12	0.4	1.3	1.3	21.0	1.0	1.6	1.8	29.8		
16	0.4	1.3	1.3	23.3	1.0	1.6	1.9	32.3		
20	0.4	1.3	1.4	25.6	1.2	2.0	2.0	36.0		
24	0.4	1.5	1.5	28.0	1.2	2.0	2.0	38.4		
1.0 mm²										
2	0.4	1.1	1.1	13.4	0.8	1.25	1.6	20.7		
4	0.4	1.2	1.1	15.3	1.0	1.25	1.7	23.2		
6	0.4	1.2	1.2	17.9	1.0	1.6	1.7	26.5		
8	0.4	1.2	1.2	18.8	1.0	1.6	1.7	27.4		
10	0.4	1.3	1.3	21.4	1.0	1.6	1.8	30.2		
12	0.4	1.3	1.3	22.2	1.0	1.6	1.9	31.2		
16	0.4	1.3	1.4	24.9	1.0	1.6	1.9	33.9		
20	0.4	1.5	1.4	27.5	1.2	2.0	2.0	37.9		
24	0.4	1.5	1.5	29.7	1.2	2.0	2.1	40.3		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl PiMF

1.3 mm²										
2	0.4	1.1	1.1	14.0	0.8	1.25	1.6	21.3		
4	0.4	1.2	1.1	16.0	1.0	1.6	1.7	24.6		
6	0.4	1.2	1.2	18.8	1.0	1.6	1.7	27.4		
8	0.4	1.2	1.2	19.8	1.0	1.6	1.8	28.6		
10	0.4	1.3	1.3	22.5	1.0	1.6	1.9	31.5		
12	0.4	1.3	1.3	23.5	1.0	1.6	1.9	32.5		
16	0.4	1.5	1.4	26.7	1.2	2.0	2.0	37.1		
20	0.4	1.5	1.5	29.3	1.2	2.0	2.1	39.9		
24	0.4	1.7	1.6	32.1	1.2	2.0	2.2	42.9		
1.5 mm²										
2	0.5	1.2	1.1	15.3	1.0	1.25	1.7	23.2		
4	0.5	1.2	1.2	17.5	1.0	1.6	1.7	26.1		
6	0.5	1.3	1.3	20.8	1.0	1.6	1.8	29.6		
8	0.5	1.3	1.3	22.0	1.0	1.6	1.9	31.0		
10	0.5	1.3	1.4	24.8	1.0	1.6	1.9	33.8		
12	0.5	1.5	1.4	26.2	1.2	2.0	2.0	36.6		
16	0.5	1.5	1.5	29.4	1.2	2.0	2.1	40.0		
20	0.5	1.7	1.6	32.7	1.2	2.0	2.2	43.5		
24	0.5	1.7	1.6	35.1	1.2	2.5	2.3	47.1		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

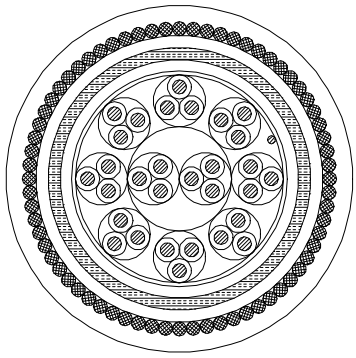
Metal sheath..... lead

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
15 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
M	sheath of lead
SWA	steel wire armour
-fl	reduced flame propagation

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	RT of lead sheath nom. (mm)	Dia. over lead sheath approx. (mm)	RT of bedding nom. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²										
1	0.4	0.8	1.1	8.4	0.8	0.9	1.4	14.6		
2	0.4	0.9	1.1	11.5	0.8	1.25	1.5	18.6		
4	0.4	1.1	1.1	13.1	0.8	1.25	1.6	20.4		
6	0.4	1.2	1.1	15.7	1.0	1.6	1.7	24.3		
8	0.4	1.2	1.1	16.6	1.0	1.6	1.7	25.2		
10	0.4	1.2	1.2	18.5	1.0	1.6	1.7	27.1		
12	0.4	1.2	1.2	19.0	1.0	1.6	1.8	27.8		
16	0.4	1.3	1.3	21.4	1.0	1.6	1.8	30.2		
20	0.4	1.3	1.3	23.2	1.0	1.6	1.9	32.2		
24	0.4	1.3	1.4	25.0	1.0	1.6	1.9	34.0		
0.75 mm²										
1	0.4	0.8	1.1	8.8	0.8	0.9	1.4	15.0		
2	0.4	1.1	1.1	12.7	0.8	1.25	1.6	20.0		
4	0.4	1.1	1.1	14.1	0.8	1.25	1.6	21.4		
6	0.4	1.2	1.2	17.2	1.0	1.6	1.7	25.8		
8	0.4	1.2	1.2	18.1	1.0	1.6	1.7	26.7		
10	0.4	1.3	1.2	20.2	1.0	1.6	1.8	29.0		
12	0.4	1.3	1.3	21.0	1.0	1.6	1.8	29.8		
16	0.4	1.3	1.3	23.3	1.0	1.6	1.9	32.3		
20	0.4	1.3	1.4	25.5	1.2	1.6	1.9	34.9		
24	0.4	1.5	1.5	28.0	1.2	2.0	2.0	38.4		
1.0 mm²										
1	0.4	0.8	1.1	9.2	0.8	0.9	1.4	15.4		
2	0.4	1.1	1.1	13.4	0.8	1.25	1.6	20.7		
4	0.4	1.2	1.1	15.1	0.8	1.25	1.6	22.4		
6	0.4	1.2	1.2	18.2	1.0	1.6	1.7	26.8		
8	0.4	1.2	1.2	19.3	1.0	1.6	1.8	28.1		
10	0.4	1.3	1.3	21.8	1.0	1.6	1.8	30.6		
12	0.4	1.3	1.3	22.4	1.0	1.6	1.9	31.4		
16	0.4	1.3	1.4	25.1	1.0	1.6	1.9	34.1		
20	0.4	1.5	1.5	27.9	1.2	2.0	2.0	38.3		
24	0.4	1.5	1.5	29.9	1.2	2.0	2.1	40.5		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl

1.3 mm²										
1	0.4	0.8	1.1	9.6	0.8	0.9	1.5	16.0		
2	0.4	1.1	1.1	14.1	0.8	1.25	1.6	21.4		
4	0.4	1.2	1.1	15.9	1.0	1.6	1.7	24.5		
6	0.4	1.2	1.2	19.3	1.0	1.6	1.8	28.1		
8	0.4	1.3	1.3	20.9	1.0	1.6	1.8	29.7		
10	0.4	1.3	1.3	23.1	1.0	1.6	1.9	32.1		
12	0.4	1.3	1.3	23.8	1.0	1.6	1.9	32.8		
16	0.4	1.5	1.4	27.1	1.2	2.0	2.0	37.5		
20	0.4	1.5	1.5	29.7	1.2	2.0	2.1	40.3		
24	0.4	1.7	1.6	32.5	1.2	2.0	2.2	43.3		
1.5 mm²										
1	0.5	0.9	1.1	10.5	0.8	0.9	1.5	16.9		
2	0.5	1.2	1.1	15.5	1.0	1.6	1.7	24.1		
4	0.5	1.2	1.2	17.6	1.0	1.6	1.7	26.2		
6	0.5	1.3	1.3	21.6	1.0	1.6	1.8	30.4		
8	0.5	1.3	1.3	22.9	1.0	1.6	1.9	31.9		
10	0.5	1.5	1.4	26.1	1.2	2.0	2.0	36.5		
12	0.5	1.5	1.4	26.9	1.2	2.0	2.0	37.3		
16	0.5	1.5	1.5	30.2	1.2	2.0	2.1	40.8		
20	0.5	1.7	1.6	33.6	1.2	2.0	2.2	44.4		
24	0.5	1.7	1.7	36.2	1.4	2.5	2.3	48.6		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...)

Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... polyvinyl chloride PVC, black

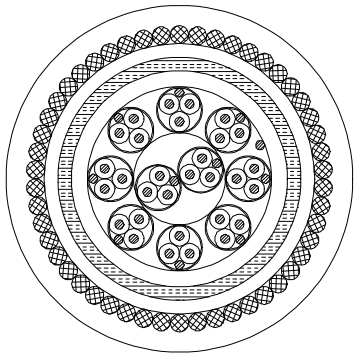
Metal sheath..... lead

Bedding..... polyvinyl chloride PVC, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V IEC 60332 PART 3**



Technical Data

Flame retardancy: IEC 60332-1

Flame propagation: IEC 60332-3 cat. C

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
15 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(St)	collective screen
Y	bedding & outer sheath of PVC
M	sheath of lead
SWA	steel wire armour
-fl	reduced flame propagation
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	RT of lead sheath nom. (mm)	Dia. over lead sheath approx. (mm)	RT of bedding nom. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²										
2	0.4	1.1	1.1	13.0	0.8	1.25	1.6	20.3		
4	0.4	1.1	1.1	14.4	0.8	1.25	1.6	21.7		
6	0.4	1.2	1.2	17.6	1.0	1.6	1.7	26.2		
8	0.4	1.2	1.2	18.6	1.0	1.6	1.7	27.2		
10	0.4	1.3	1.3	20.9	1.0	1.6	1.8	29.7		
12	0.4	1.3	1.3	21.5	1.0	1.6	1.8	30.3		
16	0.4	1.3	1.3	23.9	1.0	1.6	1.9	32.9		
20	0.4	1.5	1.4	26.6	1.2	2.0	2.0	37.0		
24	0.4	1.5	1.5	28.7	1.2	2.0	2.0	39.1		
0.75 mm²										
2	0.4	1.1	1.1	13.8	0.8	1.25	1.6	21.1		
4	0.4	1.2	1.1	15.6	1.0	1.6	1.7	24.2		
6	0.4	1.2	1.2	18.8	1.0	1.6	1.7	27.4		
8	0.4	1.2	1.2	19.9	1.0	1.6	1.8	28.7		
10	0.4	1.3	1.3	22.5	1.0	1.6	1.9	31.5		
12	0.4	1.3	1.3	23.2	1.0	1.6	1.9	32.2		
16	0.4	1.5	1.4	26.4	1.2	2.0	2.0	36.8		
20	0.4	1.5	1.5	28.9	1.2	2.0	2.0	39.3		
24	0.4	1.5	1.5	31.0	1.2	2.0	2.1	41.6		
1.0 mm²										
2	0.4	1.2	1.1	14.7	0.8	1.25	1.6	22.0		
4	0.4	1.2	1.1	16.4	1.0	1.6	1.7	25.0		
6	0.4	1.2	1.2	19.9	1.0	1.6	1.8	28.7		
8	0.4	1.3	1.3	21.5	1.0	1.6	1.8	30.3		
10	0.4	1.3	1.3	23.9	1.0	1.6	1.9	32.9		
12	0.4	1.3	1.4	24.8	1.0	1.6	1.9	33.8		
16	0.4	1.5	1.5	28.2	1.2	2.0	2.0	38.6		
20	0.4	1.5	1.5	30.7	1.2	2.0	2.1	41.3		
24	0.4	1.7	1.6	33.6	1.2	2.0	2.2	44.4		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual & Collective Screen, Lead Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(St)YMYSWAY-fl TiMF

1.3 mm²										
2	0.4	1.2	1.1	15.4	1.0	1.25	1.7	23.3		
4	0.4	1.2	1.2	17.4	1.0	1.6	1.7	26.0		
6	0.4	1.3	1.3	21.4	1.0	1.6	1.8	30.2		
8	0.4	1.3	1.3	22.7	1.0	1.6	1.9	31.7		
10	0.4	1.3	1.4	25.4	1.2	1.6	1.9	34.8		
12	0.4	1.5	1.4	26.6	1.2	2.0	2.0	37.0		
16	0.4	1.5	1.5	29.8	1.2	2.0	2.1	40.4		
20	0.4	1.7	1.6	33.1	1.2	2.0	2.2	43.9		
24	0.4	1.7	1.7	35.8	1.4	2.5	2.3	48.2		
1.5 mm²										
2	0.5	1.2	1.1	16.6	1.0	1.6	1.7	25.2		
4	0.5	1.2	1.2	18.9	1.0	1.6	1.7	27.5		
6	0.5	1.3	1.3	23.3	1.0	1.6	1.9	32.3		
8	0.5	1.3	1.4	24.9	1.0	1.6	1.9	33.9		
10	0.5	1.5	1.5	28.4	1.2	2.0	2.0	38.8		
12	0.5	1.5	1.5	29.3	1.2	2.0	2.1	39.9		
16	0.5	1.7	1.6	33.3	1.2	2.0	2.2	44.1		
20	0.5	1.7	1.7	36.6	1.4	2.5	2.3	49.0		
24	0.5	2.0	1.8	40.1	1.4	2.5	2.4	52.7		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

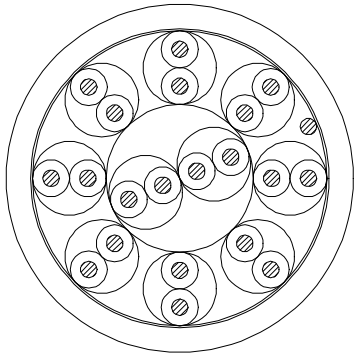
Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...) for multipairs

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Smoke density:	IEC 61034-1 and 2
Degree of acidity of gases:	IEC 60754-2
Amount of halogen acid gas:	**0%** (IEC 60754-1)
Outer sheath:	
Limiting Oxygen Index (LOI):	**min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 250°C** (ISO 4589-3)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +90°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	outer sheath of LSZH

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.0	5.7		
2	0.4	1.0	8.1		
4	0.4	1.0	9.3		
6	0.4	1.0	11.0		
8	0.4	1.0	11.7		
10	0.4	1.2	13.6		
12	0.4	1.2	14.2		
16	0.4	1.2	16.0		
20	0.4	1.4	17.9		
24	0.4	1.4	19.3		
0.75 mm²					
1	0.4	1.0	6.1		
2	0.4	1.0	8.8		
4	0.4	1.0	10.1		
6	0.4	1.0	12.0		
8	0.4	1.2	13.2		
10	0.4	1.2	14.9		
12	0.4	1.2	15.6		
16	0.4	1.4	18.0		
20	0.4	1.4	19.8		
24	0.4	1.4	21.3		
1.0 mm²					
1	0.4	1.0	6.5		
2	0.4	1.0	9.4		
4	0.4	1.0	10.9		
6	0.4	1.2	13.4		
8	0.4	1.2	14.2		
10	0.4	1.2	16.1		
12	0.4	1.2	16.8		
16	0.4	1.4	19.4		
20	0.4	1.4	21.3		
24	0.4	1.6	23.5		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H

1.3 mm²					
1	0.4	1.0	6.8		
2	0.4	1.0	10.0		
4	0.4	1.0	11.6		
6	0.4	1.2	14.3		
8	0.4	1.2	15.2		
10	0.4	1.2	17.2		
12	0.4	1.4	18.4		
16	0.4	1.4	20.8		
20	0.4	1.6	23.3		
24	0.4	1.6	25.3		
1.5 mm²					
1	0.5	1.0	7.5		
2	0.5	1.0	11.1		
4	0.5	1.2	13.3		
6	0.5	1.2	15.9		
8	0.5	1.2	16.9		
10	0.5	1.4	19.7		
12	0.5	1.4	20.6		
16	0.5	1.6	23.8		
20	0.5	1.6	26.2		
24	0.5	1.8	28.7		

RT = radial thickness

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Instrumentation Cable

KERPEN - STANDARD

Zero Halogen & Flame Retardant

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...)

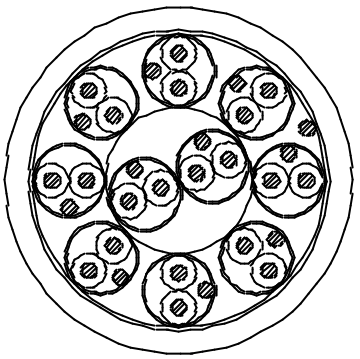
Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	outer sheath of LSZH
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

KERPEN - STANDARD

Zero Halogen & Flame Retardant

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
2	0.4	1.0	9.1		
4	0.4	1.0	10.5		
6	0.4	1.2	12.9		
8	0.4	1.2	13.7		
10	0.4	1.2	15.5		
12	0.4	1.2	16.2		
16	0.4	1.4	18.7		
20	0.4	1.4	20.5		
24	0.4	1.4	22.2		
0.75 mm²					
2	0.4	1.0	9.8		
4	0.4	1.0	11.4		
6	0.4	1.2	14.0		
8	0.4	1.2	14.9		
10	0.4	1.2	16.8		
12	0.4	1.4	18.0		
16	0.4	1.4	20.3		
20	0.4	1.4	22.4		
24	0.4	1.6	24.7		
1.0 mm²					
2	0.4	1.0	10.4		
4	0.4	1.2	12.5		
6	0.4	1.2	14.9		
8	0.4	1.2	15.8		
10	0.4	1.4	18.4		
12	0.4	1.4	19.2		
16	0.4	1.4	21.7		
20	0.4	1.6	24.4		
24	0.4	1.6	26.4		

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Instrumentation Cable

KERPEN - STANDARD

Zero Halogen & Flame Retardant

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H PiMF

1.3 mm²					
2	0.4	1.0	11.0		
4	0.4	1.2	13.2		
6	0.4	1.2	15.8		
8	0.4	1.2	16.8		
10	0.4	1.4	19.5		
12	0.4	1.4	20.5		
16	0.4	1.6	23.5		
20	0.4	1.6	26.0		
24	0.4	1.6	28.2		
1.5 mm²					
2	0.5	1.2	12.5		
4	0.5	1.2	14.5		
6	0.5	1.2	17.4		
8	0.5	1.4	19.0		
10	0.5	1.4	21.6		
12	0.5	1.4	22.6		
16	0.5	1.6	26.1		
20	0.5	1.8	29.2		
24	0.5	1.8	31.6		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

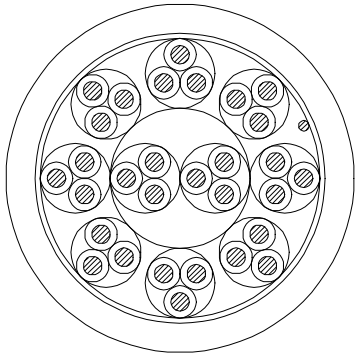
Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	outer sheath of LSZH

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, LSZH-Sheath
RE-2X(St)H

90°C / 300 V

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
1	0.4	1.0	6.0		
2	0.4	1.0	8.9		
4	0.4	1.0	10.1		
6	0.4	1.2	12.9		
8	0.4	1.2	13.8		
10	0.4	1.2	15.5		
12	0.4	1.2	16.0		
16	0.4	1.4	18.4		
20	0.4	1.4	20.2		
24	0.4	1.4	21.8		
0.75 mm²					
1	0.4	1.0	6.4		
2	0.4	1.0	9.7		
4	0.4	1.0	11.1		
6	0.4	1.2	14.2		
8	0.4	1.2	15.1		
10	0.4	1.2	17.0		
12	0.4	1.4	18.0		
16	0.4	1.4	20.3		
20	0.4	1.4	22.3		
24	0.4	1.6	24.7		
1.0 mm²					
1	0.4	1.0	6.8		
2	0.4	1.0	10.4		
4	0.4	1.0	11.9		
6	0.4	1.2	15.2		
8	0.4	1.2	16.3		
10	0.4	1.4	18.8		
12	0.4	1.4	19.4		
16	0.4	1.4	21.9		
20	0.4	1.6	24.6		
24	0.4	1.6	26.6		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H

1.3 mm²					
1	0.4	1.0	7.2		
2	0.4	1.0	11.1		
4	0.4	1.2	13.1		
6	0.4	1.2	16.3		
8	0.4	1.4	17.9		
10	0.4	1.4	20.1		
12	0.4	1.4	20.8		
16	0.4	1.6	24.0		
20	0.4	1.6	26.4		
24	0.4	1.8	29.0		
1.5 mm²					
1	0.5	1.0	7.9		
2	0.5	1.2	12.7		
4	0.5	1.2	14.6		
6	0.5	1.4	18.6		
8	0.5	1.4	19.9		
10	0.5	1.4	22.5		
12	0.5	1.6	23.8		
16	0.5	1.6	26.9		
20	0.5	1.8	30.1		
24	0.5	1.8	32.5		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Not recommended for direct burial. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...)

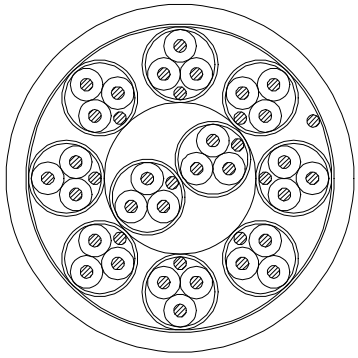
Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
7.5 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	outer sheath of LSZH
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²					
2	0.4	1.0	10.0		
4	0.4	1.0	11.4		
6	0.4	1.2	14.6		
8	0.4	1.2	15.6		
10	0.4	1.4	17.9		
12	0.4	1.4	18.5		
16	0.4	1.4	20.9		
20	0.4	1.6	23.4		
24	0.4	1.6	25.4		
0.75 mm²					
2	0.4	1.0	10.8		
4	0.4	1.2	12.8		
6	0.4	1.2	15.8		
8	0.4	1.2	16.9		
10	0.4	1.4	19.5		
12	0.4	1.4	20.2		
16	0.4	1.4	22.8		
20	0.4	1.6	25.6		
24	0.4	1.6	27.7		
1.0 mm²					
2	0.4	1.0	11.5		
4	0.4	1.2	13.6		
6	0.4	1.2	16.9		
8	0.4	1.4	18.5		
10	0.4	1.4	20.9		
12	0.4	1.4	21.6		
16	0.4	1.6	24.9		
20	0.4	1.6	27.4		
24	0.4	1.8	30.1		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, LSZH-Sheath **90°C / 300 V**

RE-2X(St)H TiMF

1.3 mm²					
2	0.4	1.2	12.6		
4	0.4	1.2	14.4		
6	0.4	1.4	18.4		
8	0.4	1.4	19.7		
10	0.4	1.4	22.2		
12	0.4	1.6	23.4		
16	0.4	1.6	26.5		
20	0.4	1.8	29.6		
24	0.4	1.8	32.1		
1.5 mm²					
2	0.5	1.2	13.8		
4	0.5	1.2	15.9		
6	0.5	1.4	20.3		
8	0.5	1.4	21.7		
10	0.5	1.6	25.1		
12	0.5	1.6	26.0		
16	0.5	1.8	29.8		
20	0.5	1.8	32.9		
24	0.5	2.0	36.0		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national regulations. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...) for multipairs

Wrapping..... at least 1 layer of plastic tape

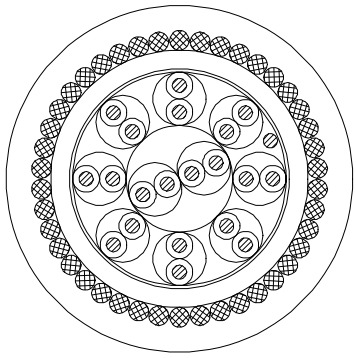
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... zero halogen flame retardant compound LSZH, black

Armour..... galvanized round steel wires

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy:	IEC 60332-1
Flame propagation:	IEC 60332-3 cat. C
Smoke density:	IEC 61034-1 and 2
Degree of acidity of gases:	IEC 60754-2
Amount of halogen acid gas:	**0%** (IEC 60754-1)
Outer sheath:	
Limiting Oxygen Index (LOI):	**min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT):	**min. 250°C** (ISO 4589-3)
Oil resistance:	(ICEA S-82-552)

Temperature range:
-30°C up to +90°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	bedding & outer sheath of LSZH
SWA	steel wire armour

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	0.8	5.3	0.9	1.3	9.7		
2	0.4	0.9	7.9	0.9	1.4	12.5		
4	0.4	0.9	9.1	0.9	1.4	13.7		
6	0.4	1.1	11.2	0.9	1.5	16.0		
8	0.4	1.1	11.9	0.9	1.5	16.7		
10	0.4	1.2	13.6	1.25	1.6	19.3		
12	0.4	1.2	14.2	1.25	1.6	19.9		
16	0.4	1.2	16.0	1.25	1.6	21.7		
20	0.4	1.3	17.7	1.6	1.7	24.3		
24	0.4	1.3	19.1	1.6	1.7	25.7		
0.75 mm²								
1	0.4	0.8	5.7	0.9	1.3	10.1		
2	0.4	0.9	8.6	0.9	1.4	13.2		
4	0.4	1.1	10.3	0.9	1.4	14.9		
6	0.4	1.1	12.2	0.9	1.5	17.0		
8	0.4	1.2	13.2	1.25	1.5	18.7		
10	0.4	1.2	14.9	1.25	1.6	20.6		
12	0.4	1.2	15.6	1.25	1.6	21.3		
16	0.4	1.3	17.8	1.6	1.7	24.4		
20	0.4	1.3	19.6	1.6	1.7	26.2		
24	0.4	1.3	21.1	1.6	1.8	27.9		
1.0 mm²								
1	0.4	0.8	6.1	0.9	1.3	10.5		
2	0.4	0.9	9.2	0.9	1.4	13.8		
4	0.4	1.1	11.1	0.9	1.5	15.9		
6	0.4	1.2	13.4	1.25	1.5	18.9		
8	0.4	1.2	14.2	1.25	1.6	19.9		
10	0.4	1.2	16.1	1.25	1.6	21.8		
12	0.4	1.2	16.8	1.25	1.6	22.5		
16	0.4	1.3	19.2	1.6	1.7	25.8		
20	0.4	1.3	21.1	1.6	1.8	27.9		
24	0.4	1.5	23.3	1.6	1.8	30.1		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

1.3 mm²								
1	0.4	0.8	6.4	0.9	1.3	10.8		
2	0.4	0.9	9.8	0.9	1.4	14.4		
4	0.4	1.1	11.8	0.9	1.5	16.6		
6	0.4	1.2	14.3	1.25	1.6	20.0		
8	0.4	1.2	15.2	1.25	1.6	20.9		
10	0.4	1.2	17.2	1.25	1.7	23.1		
12	0.4	1.3	18.2	1.6	1.7	24.8		
16	0.4	1.3	20.6	1.6	1.7	27.2		
20	0.4	1.3	22.7	1.6	1.8	29.5		
24	0.4	1.5	25.1	1.6	1.9	32.1		
1.5 mm²								
1	0.5	0.8	7.1	0.9	1.3	11.5		
2	0.5	1.1	11.3	0.9	1.5	16.1		
4	0.5	1.2	13.3	1.25	1.5	18.8		
6	0.5	1.2	15.9	1.25	1.6	21.6		
8	0.5	1.2	16.9	1.25	1.7	22.8		
10	0.5	1.3	19.5	1.6	1.7	26.1		
12	0.5	1.3	20.4	1.6	1.7	27.0		
16	0.5	1.5	23.6	1.6	1.8	30.4		
20	0.5	1.5	26.0	1.6	1.9	33.0		
24	0.5	1.5	28.1	2.0	2.0	36.1		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national regulations. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white; continuously numbered on cores (1-1, 2-2, ...)

Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

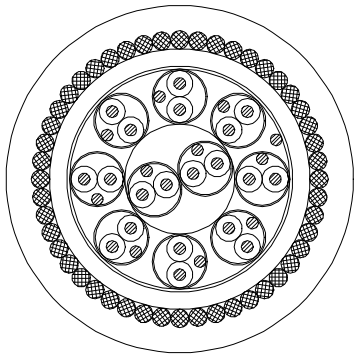
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... zero halogen flame retardant compound LSZH, black

Armour..... galvanized round steel wires

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	bedding & outer sheath of LSZH
SWA	steel wire armour
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	0.9	8.9	0.9	1.4	13.5		
4	0.4	1.1	10.7	0.9	1.4	15.3		
6	0.4	1.2	12.9	1.25	1.5	18.4		
8	0.4	1.2	13.7	1.25	1.6	19.4		
10	0.4	1.2	15.5	1.25	1.6	21.2		
12	0.4	1.2	16.2	1.25	1.6	21.9		
16	0.4	1.3	18.5	1.6	1.7	25.1		
20	0.4	1.3	20.3	1.6	1.7	26.9		
24	0.4	1.3	22.0	1.6	1.8	28.8		
0.75 mm²								
2	0.4	0.9	9.6	0.9	1.4	14.2		
4	0.4	1.1	11.6	0.9	1.5	16.4		
6	0.4	1.2	14.0	1.25	1.6	19.7		
8	0.4	1.2	14.9	1.25	1.6	20.6		
10	0.4	1.2	16.8	1.25	1.6	22.5		
12	0.4	1.3	17.8	1.6	1.7	24.4		
16	0.4	1.3	20.1	1.6	1.7	26.7		
20	0.4	1.3	22.2	1.6	1.8	29.0		
24	0.4	1.5	24.5	1.6	1.9	31.5		
1.0 mm²								
2	0.4	1.1	10.6	0.9	1.4	15.2		
4	0.4	1.2	12.5	1.25	1.5	18.0		
6	0.4	1.2	14.9	1.25	1.6	20.6		
8	0.4	1.2	15.8	1.25	1.6	21.5		
10	0.4	1.3	18.2	1.6	1.7	24.8		
12	0.4	1.3	19.0	1.6	1.7	25.6		
16	0.4	1.3	21.5	1.6	1.8	28.3		
20	0.4	1.5	24.2	1.6	1.9	31.2		
24	0.4	1.5	26.2	1.6	1.9	33.2		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Pair, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH PiMF

1.3 mm²								
2	0.4	1.1	11.2	0.9	1.5	16.0		
4	0.4	1.2	13.2	1.25	1.5	18.7		
6	0.4	1.2	15.8	1.25	1.6	21.5		
8	0.4	1.2	16.8	1.25	1.6	22.5		
10	0.4	1.3	19.3	1.6	1.7	25.9		
12	0.4	1.3	20.3	1.6	1.7	26.9		
16	0.4	1.5	23.3	1.6	1.8	30.1		
20	0.4	1.5	25.8	1.6	1.9	32.8		
24	0.4	1.5	28.0	2.0	2.0	36.0		
1.5 mm²								
2	0.5	1.2	12.5	1.25	1.5	18.0		
4	0.5	1.2	14.5	1.25	1.6	20.2		
6	0.5	1.2	17.4	1.25	1.7	23.3		
8	0.5	1.3	18.8	1.6	1.7	25.4		
10	0.5	1.3	21.4	1.6	1.8	28.2		
12	0.5	1.3	22.4	1.6	1.8	29.2		
16	0.5	1.5	25.9	1.6	1.9	32.9		
20	0.5	1.7	29.0	2.0	2.0	37.0		
24	0.5	1.7	31.4	2.0	2.0	39.4		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national regulations. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

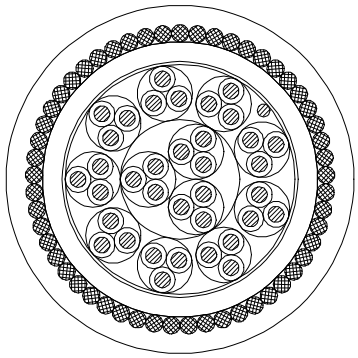
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... zero halogen flame retardant compound LSZH, black

Armour..... galvanized round steel wires

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	bedding & outer sheath of LSZH
SWA	steel wire armour

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	0.8	5.6	0.9	1.3	10.0		
2	0.4	0.9	8.7	0.9	1.4	13.3		
4	0.4	1.1	10.3	0.9	1.4	14.9		
6	0.4	1.2	12.9	1.25	1.5	18.4		
8	0.4	1.2	13.8	1.25	1.6	19.5		
10	0.4	1.2	15.5	1.25	1.6	21.2		
12	0.4	1.2	16.0	1.25	1.6	21.7		
16	0.4	1.3	18.2	1.6	1.7	24.8		
20	0.4	1.3	20.0	1.6	1.7	26.6		
24	0.4	1.3	21.6	1.6	1.8	28.4		
0.75 mm²								
1	0.4	0.8	6.0	0.9	1.3	10.4		
2	0.4	0.9	9.5	0.9	1.4	14.1		
4	0.4	1.1	11.3	0.9	1.5	16.1		
6	0.4	1.2	14.2	1.25	1.6	19.9		
8	0.4	1.2	15.1	1.25	1.6	20.8		
10	0.4	1.2	17.0	1.25	1.7	22.9		
12	0.4	1.3	17.8	1.6	1.7	24.4		
16	0.4	1.3	20.1	1.6	1.7	26.7		
20	0.4	1.3	22.1	1.6	1.8	28.9		
24	0.4	1.5	24.5	1.6	1.9	31.5		
1.0 mm²								
1	0.4	0.8	6.4	0.9	1.3	10.8		
2	0.4	1.1	10.6	0.9	1.4	15.2		
4	0.4	1.1	12.1	0.9	1.5	16.9		
6	0.4	1.2	15.2	1.25	1.6	20.9		
8	0.4	1.2	16.3	1.25	1.6	22.0		
10	0.4	1.3	18.6	1.6	1.7	25.2		
12	0.4	1.3	19.2	1.6	1.7	25.8		
16	0.4	1.3	21.7	1.6	1.8	28.5		
20	0.4	1.5	24.4	1.6	1.9	31.4		
24	0.4	1.5	26.4	1.6	1.9	33.4		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, XLPE-Insulation, Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH

1.3 mm²								
1	0.4	0.8	6.8	0.9	1.3	11.2		
2	0.4	1.1	11.3	0.9	1.5	16.1		
4	0.4	1.2	13.1	1.25	1.5	18.6		
6	0.4	1.2	16.3	1.25	1.6	22.0		
8	0.4	1.3	17.7	1.6	1.7	24.3		
10	0.4	1.3	19.9	1.6	1.7	26.5		
12	0.4	1.3	20.6	1.6	1.7	27.2		
16	0.4	1.5	23.8	1.6	1.8	30.6		
20	0.4	1.5	26.2	1.6	1.9	33.2		
24	0.4	1.7	28.8	2.0	2.0	36.8		
1.5 mm²								
1	0.5	0.8	7.5	0.9	1.4	12.1		
2	0.5	1.2	12.7	1.25	1.5	18.2		
4	0.5	1.2	14.6	1.25	1.6	20.3		
6	0.5	1.3	18.4	1.6	1.7	25.0		
8	0.5	1.3	19.7	1.6	1.7	26.3		
10	0.5	1.3	22.3	1.6	1.8	29.1		
12	0.5	1.5	23.6	1.6	1.8	30.4		
16	0.5	1.5	26.7	1.6	1.9	33.7		
20	0.5	1.7	29.9	2.0	2.0	37.9		
24	0.5	1.7	32.3	2.0	2.1	40.5		

RT = radial thickness

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial acc. to national regulations. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... cross-linked polyethylene XLPE

Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ...)

Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

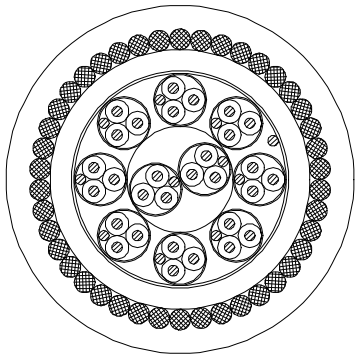
Collective screen... 24 µm aluminium / PETP tape over 7-stranded tinned copper drain wire, 0.5 mm²

Bedding..... zero halogen flame retardant compound LSZH, black

Armour..... galvanized round steel wires

Outer sheath..... zero halogen flame retardant compound LSZH, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V LSZH**



Technical Data

Flame retardancy: IEC 60332-1
Flame propagation: IEC 60332-3 cat. C
Smoke density: IEC 61034-1 and 2
Degree of acidity of gases: IEC 60754-2
Amount of halogen acid gas: **0%** (IEC 60754-1)

Outer sheath:
Limiting Oxygen Index (LOI): **min. 35%** (IEC 60332-3 annex B)
Flame Temperature (FT): **min. 250°C** (ISO 4589-3)
Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +90°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2X	insulation of XLPE
(St)	collective screen
H	bedding & outer sheath of LSZH
SWA	steel wire armour
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of bedding nom. (mm)	Dia. over bedding approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	0.9	9.8	0.9	1.4	14.4		
4	0.4	1.1	11.6	0.9	1.5	16.4		
6	0.4	1.2	14.6	1.25	1.6	20.3		
8	0.4	1.2	15.6	1.25	1.6	21.3		
10	0.4	1.3	17.7	1.6	1.7	24.3		
12	0.4	1.3	18.3	1.6	1.7	24.9		
16	0.4	1.3	20.7	1.6	1.7	27.3		
20	0.4	1.3	22.8	1.6	1.8	29.6		
24	0.4	1.5	25.2	1.6	1.9	32.2		
0.75 mm²								
2	0.4	1.1	11.0	0.9	1.5	15.8		
4	0.4	1.2	12.8	1.25	1.5	18.3		
6	0.4	1.2	15.8	1.25	1.6	21.5		
8	0.4	1.2	16.9	1.25	1.7	22.8		
10	0.4	1.3	19.3	1.6	1.7	25.9		
12	0.4	1.3	20.0	1.6	1.7	26.6		
16	0.4	1.3	22.6	1.6	1.8	29.4		
20	0.4	1.5	25.4	1.6	1.9	32.4		
24	0.4	1.5	27.5	1.6	1.9	34.5		
1.0 mm²								
2	0.4	1.1	11.7	0.9	1.5	16.5		
4	0.4	1.2	13.6	1.25	1.6	19.3		
6	0.4	1.2	16.9	1.25	1.7	22.8		
8	0.4	1.3	18.3	1.6	1.7	24.9		
10	0.4	1.3	20.7	1.6	1.7	27.3		
12	0.4	1.3	21.4	1.6	1.8	28.2		
16	0.4	1.5	24.7	1.6	1.9	31.7		
20	0.4	1.5	27.2	1.6	1.9	34.2		
24	0.4	1.7	29.9	2.0	2.0	37.9		

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Instrumentation Cable

Zero Halogen & Flame Retardant

KERPEN - STANDARD

Multi-Triple, XLPE-Insulation, Individual & Collective Screen, Armour, LSZH-Sheath **90°C / 300 V**

RE-2X(St)HSWAH TiMF

1.3 mm²								
2	0.4	1.2	12.6	1.25	1.5	18.1		
4	0.4	1.2	14.4	1.25	1.6	20.1		
6	0.4	1.3	18.2	1.6	1.7	24.8		
8	0.4	1.3	19.5	1.6	1.7	26.1		
10	0.4	1.3	22.0	1.6	1.8	28.8		
12	0.4	1.3	22.8	1.6	1.8	29.6		
16	0.4	1.5	26.3	1.6	1.9	33.3		
20	0.4	1.7	29.4	2.0	2.0	37.4		
24	0.4	1.7	31.9	2.0	2.1	40.1		
1.5 mm²								
2	0.5	1.2	13.8	1.25	1.6	19.5		
4	0.5	1.2	15.9	1.25	1.6	21.6		
6	0.5	1.3	20.1	1.6	1.7	26.7		
8	0.5	1.3	21.5	1.6	1.8	28.3		
10	0.5	1.5	24.9	1.6	1.9	31.9		
12	0.5	1.5	25.8	1.6	1.9	32.8		
16	0.5	1.7	29.6	2.0	2.0	37.6		
20	0.5	1.7	32.7	2.0	2.1	40.9		
24	0.5	2.0	36.0	2.0	2.2	44.4		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...) for multipairs

Wrapping..... at least 1 layer of plastic tape

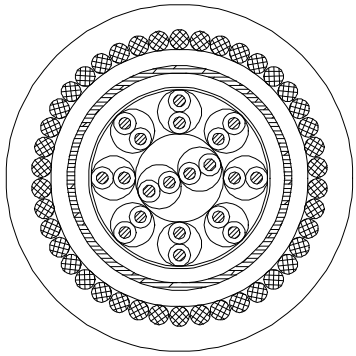
Laminated sheath.. one side plastic coated aluminium tape (thickness (min.) 0.15 mm) longitudinally applied over tinned copper drain wire (0.5 mm ²), seam sealed and bonded to the high density polyethylene HDPE-sheath, black

Protection sheath.. heat stabilized polyamide, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**



Technical Data

Flame retardancy: IEC 60332-1

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C
(during operation)
- 5°C up to +50°C
(during installation)

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(L)2Y	laminated sheath
4Y	sheath of polyamide
SWA	steel wire armour
Yfl	outer sheath of PVC flame retardant

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single pair:					120		
up to 4 pairs:					96		
above 4 pairs:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of HDPE of the lamin. sheath nom. (mm)	Dia. under armour approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	1.0	7.8	0.9	1.4	12.4		
2	0.4	1.0	10.2	0.9	1.4	14.8		
4	0.4	1.0	11.4	0.9	1.5	16.2		
6	0.4	1.0	13.3	1.25	1.5	18.8		
8	0.4	1.0	14.0	1.25	1.6	19.7		
10	0.4	1.0	15.5	1.25	1.6	21.2		
12	0.4	1.0	16.1	1.6	1.6	22.5		
16	0.4	1.0	17.9	1.6	1.7	24.5		
20	0.4	1.0	19.4	1.6	1.7	26.0		
24	0.4	1.0	20.8	1.6	1.7	27.4		
0.75 mm²								
1	0.4	1.0	8.2	0.9	1.4	12.8		
2	0.4	1.0	10.9	0.9	1.5	15.7		
4	0.4	1.0	12.2	0.9	1.5	17.0		
6	0.4	1.0	14.3	1.25	1.6	20.0		
8	0.4	1.0	15.1	1.25	1.6	20.8		
10	0.4	1.0	16.8	1.6	1.7	23.4		
12	0.4	1.0	17.5	1.6	1.7	24.1		
16	0.4	1.0	19.5	1.6	1.7	26.1		
20	0.4	1.0	21.3	1.6	1.8	28.1		
24	0.4	1.0	22.8	1.6	1.8	29.6		
1.0 mm²								
1	0.4	1.0	8.6	0.9	1.4	13.2		
2	0.4	1.0	11.5	0.9	1.5	16.3		
4	0.4	1.0	13.0	1.25	1.5	18.5		
6	0.4	1.0	15.3	1.25	1.6	21.0		
8	0.4	1.0	16.1	1.6	1.6	22.5		
10	0.4	1.0	18.0	1.6	1.7	24.6		
12	0.4	1.0	18.7	1.6	1.7	25.3		
16	0.4	1.0	20.9	1.6	1.7	27.5		
20	0.4	1.0	22.8	1.6	1.8	29.6		
24	0.4	1.0	24.6	1.6	1.9	31.6		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Pair, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

1.3 mm²								
1	0.4	1.0	8.9	0.9	1.4	13.5		
2	0.4	1.0	12.1	0.9	1.5	16.9		
4	0.4	1.0	13.9	1.25	1.6	19.6		
6	0.4	1.0	16.2	1.6	1.7	22.8		
8	0.4	1.0	17.1	1.6	1.7	23.7		
10	0.4	1.0	19.1	1.6	1.7	25.7		
12	0.4	1.0	19.9	1.6	1.7	26.5		
16	0.4	1.0	22.3	1.6	1.8	29.1		
20	0.4	1.0	24.4	1.6	1.9	31.4		
24	0.4	1.0	26.4	1.6	1.9	33.4		
1.5 mm²								
1	0.5	1.0	9.6	0.9	1.4	14.2		
2	0.5	1.0	13.4	1.25	1.5	18.9		
4	0.5	1.0	15.2	1.25	1.6	20.9		
6	0.5	1.0	17.8	1.6	1.7	24.4		
8	0.5	1.0	18.8	1.6	1.7	25.4		
10	0.5	1.0	21.2	1.6	1.8	28.0		
12	0.5	1.0	22.1	1.6	1.8	28.9		
16	0.5	1.0	24.9	1.6	1.9	31.9		
20	0.5	1.0	27.3	1.6	1.9	34.3		
24	0.5	1.0	29.4	1.6	2.0	36.6		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual Screen, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl PiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white;
continuously numbered on cores (1-1, 2-2, ...)

Pair screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

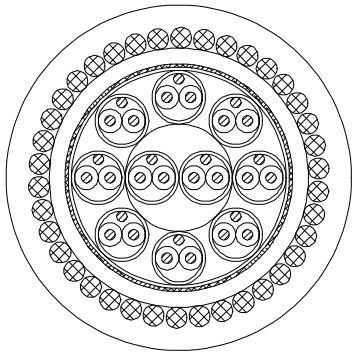
Laminated sheath.. one side plastic coated aluminium tape (thickness (min.) 0.15 mm) longitudinally applied over tinned copper drain wire (0.5 mm ²), seam sealed and bonded to the high density polyethylene HDPE-sheath, black

Protection sheath.. heat stabilized polyamide, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**



Technical Data

Flame retardancy: IEC 60332-1

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(L)2Y	laminated sheath
4Y	sheath of polyamide
SWA	steel wire armour
Yfl	outer sheath of PVC flame retardant
PiMF	pair screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual Screen, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl PiMF

Geometrical data

No. of pairs	RT of insulation nom. (mm)	RT of HDPE of the lamin. sheath nom. (mm)	Dia. under armour approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	1.0	11.2	0.9	1.5	16.0		
4	0.4	1.0	12.6	1.25	1.5	18.1		
6	0.4	1.0	14.8	1.25	1.6	20.5		
8	0.4	1.0	15.6	1.25	1.6	21.3		
10	0.4	1.0	17.4	1.6	1.7	24.0		
12	0.4	1.0	18.1	1.6	1.7	24.7		
16	0.4	1.0	20.2	1.6	1.7	26.8		
20	0.4	1.0	22.0	1.6	1.8	28.8		
24	0.4	1.0	23.7	1.6	1.8	30.5		
0.75 mm²								
2	0.4	1.0	11.9	0.9	1.5	16.7		
4	0.4	1.0	13.7	1.25	1.6	19.4		
6	0.4	1.0	15.9	1.25	1.6	21.6		
8	0.4	1.0	16.8	1.6	1.7	23.4		
10	0.4	1.0	18.7	1.6	1.7	25.3		
12	0.4	1.0	19.5	1.6	1.7	26.1		
16	0.4	1.0	21.8	1.6	1.8	28.6		
20	0.4	1.0	23.9	1.6	1.8	30.7		
24	0.4	1.0	25.8	1.6	1.9	32.8		
1.0 mm²								
2	0.4	1.0	12.5	1.25	1.5	18.0		
4	0.4	1.0	14.4	1.25	1.6	20.1		
6	0.4	1.0	16.8	1.6	1.7	23.4		
8	0.4	1.0	17.7	1.6	1.7	24.3		
10	0.4	1.0	19.9	1.6	1.7	26.5		
12	0.4	1.0	20.7	1.6	1.7	27.3		
16	0.4	1.0	23.2	1.6	1.8	30.0		
20	0.4	1.0	25.5	1.6	1.9	32.5		
24	0.4	1.0	27.5	1.6	1.9	34.5		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Pair, PE-Insulation, Individual Screen, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl PiMF

1.3 mm²								
2	0.4	1.0	13.3	1.25	1.5	18.8		
4	0.4	1.0	15.1	1.25	1.6	20.8		
6	0.4	1.0	17.7	1.6	1.7	24.3		
8	0.4	1.0	18.7	1.6	1.7	25.3		
10	0.4	1.0	21.0	1.6	1.8	27.8		
12	0.4	1.0	22.0	1.6	1.8	28.8		
16	0.4	1.0	24.6	1.6	1.9	31.6		
20	0.4	1.0	27.1	1.6	1.9	34.1		
24	0.4	1.0	29.3	1.6	2.0	36.5		
1.5 mm²								
2	0.5	1.0	14.4	1.25	1.6	20.1		
4	0.5	1.0	16.4	1.6	1.7	23.0		
6	0.5	1.0	19.3	1.6	1.7	25.9		
8	0.5	1.0	20.5	1.6	1.7	27.1		
10	0.5	1.0	23.1	1.6	1.8	29.9		
12	0.5	1.0	24.1	1.6	1.9	31.1		
16	0.5	1.0	27.2	1.6	1.9	34.2		
20	0.5	1.0	29.9	1.6	2.0	37.1		
24	0.5	1.0	32.3	2.0	2.1	40.5		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red;
continuously numbered on cores (1-1-1, 2-2-2 ...) for multitriples

Wrapping..... at least 1 layer of plastic tape

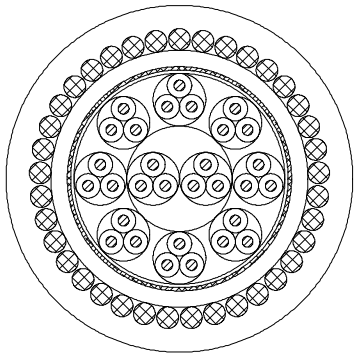
Laminated sheath.. one side plastic coated aluminium tape (thickness (min.) 0.15 mm) longitudinally applied over tinned copper drain wire (0.5 mm ²), seam sealed and bonded to the high density polyethylene HDPE-sheath, black

Protection sheath.. heat stabilized polyamide, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**



Technical Data

Flame retardancy: IEC 60332-1

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(L)2Y	laminated sheath
4Y	sheath of polyamide
SWA	steel wire armour
Yfl	outer sheath of PVC flame retardant

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km					
single triple:					120		
up to 4 triples:					96		
above 4 triples:					80		
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of HDPE of the lamin. sheath nom. (mm)	Dia. under armour approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
1	0.4	1.0	8.1	0.9	1.4	12.7		
2	0.4	1.0	11.0	0.9	1.5	15.8		
4	0.4	1.0	12.2	0.9	1.5	17.0		
6	0.4	1.0	14.8	1.25	1.6	20.5		
8	0.4	1.0	15.7	1.25	1.6	21.4		
10	0.4	1.0	17.4	1.6	1.7	24.0		
12	0.4	1.0	17.9	1.6	1.7	24.5		
16	0.4	1.0	19.9	1.6	1.7	26.5		
20	0.4	1.0	21.7	1.6	1.8	28.5		
24	0.4	1.0	23.3	1.6	1.8	30.1		
0.75 mm²								
1	0.4	1.0	8.5	0.9	1.4	13.1		
2	0.4	1.0	11.8	0.9	1.5	16.6		
4	0.4	1.0	13.4	1.25	1.5	18.9		
6	0.4	1.0	16.1	1.6	1.6	22.5		
8	0.4	1.0	17.0	1.6	1.7	23.6		
10	0.4	1.0	18.9	1.6	1.7	25.5		
12	0.4	1.0	19.5	1.6	1.7	26.1		
16	0.4	1.0	21.8	1.6	1.8	28.6		
20	0.4	1.0	23.8	1.6	1.8	30.6		
24	0.4	1.0	25.8	1.6	1.9	32.8		
1.0 mm²								
1	0.4	1.0	8.9	0.9	1.4	13.5		
2	0.4	1.0	12.5	1.25	1.5	18.0		
4	0.4	1.0	14.2	1.25	1.6	19.9		
6	0.4	1.0	17.1	1.6	1.7	23.7		
8	0.4	1.0	18.2	1.6	1.7	24.8		
10	0.4	1.0	20.3	1.6	1.7	26.9		
12	0.4	1.0	20.9	1.6	1.7	27.5		
16	0.4	1.0	23.4	1.6	1.8	30.2		
20	0.4	1.0	25.7	1.6	1.9	32.7		
24	0.4	1.0	27.7	1.6	1.9	34.7		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Single- & Multi-Triple, PE-Insulation, ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl

1.3 mm²								
1	0.4	1.0	9.3	0.9	1.4	13.9		
2	0.4	1.0	13.4	1.25	1.5	18.9		
4	0.4	1.0	15.0	1.25	1.6	20.7		
6	0.4	1.0	18.2	1.6	1.7	24.8		
8	0.4	1.0	19.4	1.6	1.7	26.0		
10	0.4	1.0	21.6	1.6	1.8	28.4		
12	0.4	1.0	22.3	1.6	1.8	29.1		
16	0.4	1.0	25.1	1.6	1.9	32.1		
20	0.4	1.0	27.5	1.6	1.9	34.5		
24	0.4	1.0	29.7	1.6	2.0	36.9		
1.5 mm²								
1	0.5	1.0	10.0	0.9	1.4	14.6		
2	0.5	1.0	14.6	1.25	1.6	20.3		
4	0.5	1.0	16.5	1.6	1.7	23.1		
6	0.5	1.0	20.1	1.6	1.7	26.7		
8	0.5	1.0	21.4	1.6	1.8	28.2		
10	0.5	1.0	24.0	1.6	1.9	31.0		
12	0.5	1.0	24.9	1.6	1.9	31.9		
16	0.5	1.0	28.0	1.6	1.9	35.0		
20	0.5	1.0	30.8	2.0	2.0	38.8		
24	0.5	1.0	33.2	2.0	2.1	41.4		

RT = radial thickness

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual , ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl TiMF

Application

For transmission of analogue and digital signals in instrument and control systems; use in zone 1 and zone 2 group II classified areas (IEC 60079 part 14). Not allowed for direct connection to low impedance source, e.g. the public mains electricity supply.

Recommended for direct burial, especially in the presence of aggressive chemical substances. For indoor and outdoor installation in dry and wet locations on racks, in conduits.

Construction

Conductor..... plain annealed copper, stranded, sizes: 0.5 mm², 0.75 mm², 1.0 mm², 1.3 mm² or 1.5 mm²

Insulation..... polyethylene PE

Colour code..... black / white / red; continuously numbered on cores (1-1-1, 2-2-2 ...)

Triple screen..... 24 µm aluminium / PETP tape over solid tinned copper drain wire, 0.6 mm Ø, plastic tape under and above screen

Wrapping..... at least 1 layer of plastic tape

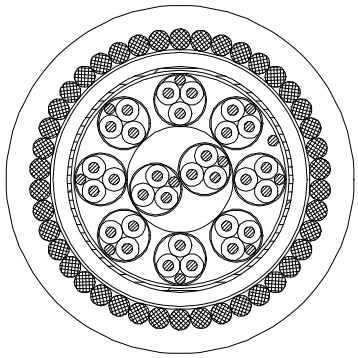
Laminated sheath.. one side plastic coated aluminium tape (thickness (min.) 0.15 mm) longitudinally applied over tinned copper drain wire (0.5 mm ²), seam sealed and bonded to the high density polyethylene HDPE-sheath, black

Protection sheath.. heat stabilized polyamide, black

Armour..... galvanized round steel wires

Outer sheath..... polyvinyl chloride PVC, black

Cable marking..... **KERPENWERK INSTRUMENTATION CABLE 300 V**



Technical Data

Flame retardancy: IEC 60332-1

Outer sheath:

Amount of halogen acid gas: **max. 17%** (IEC 60754-1)

Limiting Oxygen Index (LOI): **min. 30%** (IEC 60332-3 annex B)

Flame Temperature (FT): **min. 300°C** (ISO 4589-3)

Sunlight resistance: (UL 1581 section 1200)

Oil resistance: (ICEA S-82-552)

Temperature range:
-30°C up to +70°C *(during operation)*
- 5°C up to +50°C *(during installation)*

Min. bending radius:
10 x cable-Ø

Abbreviations

RE-	instrumentation cable
2Y	insulation of PE
(L)2Y	laminated sheath
4Y	sheath of polyamide
SWA	steel wire armour
Yfl	outer sheath of PVC flame retardant
TiMF	triple screen

Electrical data at 20°C

	Character	Unit	Values				
Conductor size	nom.	mm²	0.5	0.75	1.0	1.3	1.5
Conductor resistance	max.	Ω/km	39.2	24.6	18.4	14.2	12.6
Insulation resistance	min.	MΩ x km	5000				
Mutual capacitance at 0.8 resp. 1 kHz	max.	nF/km	120				
L/R (ratio)	max.	µH/Ω	25	25	25	40	40
Test voltage U_{rms} ***core : core***		V	2000				
U_{rms} ***core : screen***		V	1000				
Operating voltage U_{rms}	max.	V	300				

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual , ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl TiMF

Geometrical data

No. of triples	RT of insulation nom. (mm)	RT of HDPE of the lamin. sheath nom. (mm)	Dia. under armour approx. (mm)	Dia. of armour wire nom. (mm)	RT of outer sheath nom. (mm)	Overall Diameter approx. (mm)	Weight approx. (kg/km)	Part No.
0.5 mm²								
2	0.4	1.0	12.1	0.9	1.5	16.9		
4	0.4	1.0	13.7	1.25	1.6	19.4		
6	0.4	1.0	16.5	1.6	1.7	23.1		
8	0.4	1.0	17.5	1.6	1.7	24.1		
10	0.4	1.0	19.4	1.6	1.7	26.0		
12	0.4	1.0	20.0	1.6	1.7	26.6		
16	0.4	1.0	22.4	1.6	1.8	29.2		
20	0.4	1.0	24.5	1.6	1.9	31.5		
24	0.4	1.0	26.5	1.6	1.9	33.5		
0.75 mm²								
2	0.4	1.0	12.9	1.25	1.5	18.4		
4	0.4	1.0	14.7	1.25	1.6	20.4		
6	0.4	1.0	17.7	1.6	1.7	24.3		
8	0.4	1.0	18.8	1.6	1.7	25.4		
10	0.4	1.0	21.0	1.6	1.8	27.8		
12	0.4	1.0	21.7	1.6	1.8	28.5		
16	0.4	1.0	24.3	1.6	1.9	31.3		
20	0.4	1.0	26.7	1.6	1.9	33.7		
24	0.4	1.0	28.8	1.6	2.0	36.0		
1.0 mm²								
2	0.4	1.0	13.8	1.25	1.6	19.5		
4	0.4	1.0	15.5	1.25	1.6	21.2		
6	0.4	1.0	18.8	1.6	1.7	25.4		
8	0.4	1.0	20.0	1.6	1.7	26.6		
10	0.4	1.0	22.4	1.6	1.8	29.2		
12	0.4	1.0	23.1	1.6	1.8	29.9		
16	0.4	1.0	26.0	1.6	1.9	33.0		
20	0.4	1.0	28.5	1.6	2.0	35.7		
24	0.4	1.0	30.8	2.0	2.0	38.8		

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Instrumentation Cable

Flame Retardant

KERPEN - STANDARD

Multi-Triple, PE-Insulation, Individual , ALNYC-Sheath, Armour, PVC-Sheath **70°C / 300 V**

RE-2Y(L)2Y4YSWAYfl TiMF

1.3 mm²								
2	0.4	1.0	14.5	1.25	1.6	20.2		
4	0.4	1.0	16.3	1.6	1.7	22.9		
6	0.4	1.0	19.9	1.6	1.7	26.5		
8	0.4	1.0	21.2	1.6	1.8	28.0		
10	0.4	1.0	23.7	1.6	1.8	30.5		
12	0.4	1.0	24.5	1.6	1.9	31.5		
16	0.4	1.0	27.6	1.6	1.9	34.6		
20	0.4	1.0	30.3	2.0	2.0	38.3		
24	0.4	1.0	32.8	2.0	2.1	41.0		
1.5 mm²								
2	0.5	1.0	15.7	1.25	1.6	21.4		
4	0.5	1.0	17.8	1.6	1.7	24.4		
6	0.5	1.0	21.8	1.6	1.8	28.6		
8	0.5	1.0	23.2	1.6	1.8	30.0		
10	0.5	1.0	26.2	1.6	1.9	33.2		
12	0.5	1.0	27.1	1.6	1.9	34.1		
16	0.5	1.0	30.5	2.0	2.0	38.5		
20	0.5	1.0	33.6	2.0	2.1	41.8		
24	0.5	1.0	36.3	2.0	2.2	44.7		

RT = radial thickness

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